
Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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Abstract

Generating novel protein sequences that respect the statistical constraints of a family typically requires training deep generative models on thousands to millions of examples. Yet most protein families are small: the median Pfam family has fewer than 100 members, a regime where learned models overfit or collapse. We propose *stochastic attention* (SA), a training-free sampler that treats the modern Hopfield energy over a set of stored protein sequences as a Boltzmann distribution and draws samples via Langevin dynamics. The score function is given in closed form by the residual of a single softmax attention operation, eliminating the need for a trained score network, external pretraining data, or graphics processing unit (GPU) resources. Each iteration costs $\mathcal{O}(dK)$, where d is the sequence representation dimension and K is the number of stored family members — two matrix-vector products. Across eight Pfam families ($K=37\text{--}420$ sequences, $L=23\text{--}262$ residues), including antimicrobial beta defensins, SA generates sequences with low amino acid Kullback–Leibler (KL) composition divergence ($\text{KL} < 0.06$), substantial novelty (0.40–0.65), and structural plausibility confirmed by ESMFold and AlphaFold2 (mean predicted local distance difference test [pLDDT] up to 95). Strikingly, generated sequences achieve significantly *higher* template modeling scores (TM-scores) to canonical family folds than natural family members in five of seven families tested by ESMFold ($p < 0.001$), a result corroborated by AlphaFold2 cross-validation. Head-to-head comparison with profile hidden Markov models (HMMs), EvoDiff, and the multiple sequence alignment (MSA) Transformer reveals a critical distinction: learned baselines produce sequences that drift far from the family (as low as 11% sequence identity), while SA maintains 52–66% identity — staying within the functional envelope while remaining genuinely novel. Stochastic attention achieves this in seconds on a laptop, compared to hours of GPU or CPU time for pretrained alternatives, and scales stably down to $K=20$ sequences. Independent validation by ESM2-650M confirms that generated sequences are recognized as comparably protein-like to natural family members, and cross-referencing against published deep mutational scanning experiments shows that over 90% of SA-generated substitutions correspond to experimentally tolerated mutations. The critical inverse temperature β^* governing the generation–retrieval transition is predicted from principal component analysis (PCA) dimensionality alone ($R^2=0.97$), enabling fully automatic operation from a seed alignment.

Significance Statement. Most protein families have fewer than 100 known members, too few to train a deep generative model. We show that a single attention operation over a family alignment defines an energy landscape whose Boltzmann distribution can be sampled exactly via Langevin dynamics, with no training, no GPU, and no external data. The resulting method, stochastic attention, generates novel protein sequences that preserve family-level amino acid statistics and fold into the correct three-dimensional structure across eight diverse protein families, including antimicrobial peptides. A simple linear relationship between the dimensionality of the sequence representation and the critical sampling temperature enables fully automatic operation from a seed alignment, opening training-free sequence generation to the vast majority of protein families that are too small for conventional deep learning.

1 Introduction

The protein folding problem, predicting three-dimensional structure from amino acid sequence, stood as one of the grand challenges in molecular biology for half a century. AlphaFold2 [1] and RoseTTAFold [2] effectively solved it, achieving experimental-level accuracy across the proteome and earning the 2024 Nobel Prize in Chemistry. With structure prediction largely in hand, the frontier has shifted decisively toward the inverse problem: *designing* novel protein sequences with desired properties [3, 4]. This shift has catalyzed a wave of generative models for protein design, spanning structure-conditioned methods, sequence-level language models, and diffusion-based approaches. On the sequence side, profile hidden Markov models (pHMMs) [5, 6] have long captured positional emission and transition probabilities from multiple sequence alignments (MSAs), but they model only single-site statistics and miss the pairwise and higher-order couplings critical for maintaining tertiary structure. Direct coupling analysis (DCA) and Potts models [7, 8] capture pairwise epistasis, and Trinquier et al. [9] showed that autoregressive models over DCA-derived distributions can generate realistic sequences efficiently. Deep generative models have pushed further: DeepSequence [10] learns latent representations via a variational autoencoder (VAE); ProtGPT2 [11] and ProGen [12] generate sequences autoregressively from large language models pretrained on UniRef50; the MSA Transformer [13] processes entire alignments via tied row attention; EvoDiff [14] applies discrete diffusion to MSA-conditioned generation; and Tranception [15] combines autoregressive transformers with inference-time retrieval. On the structure side, ProteinMPNN [16] designs sequences conditioned on backbone coordinates via message-passing, RFdiffusion [17] generates novel backbones through denoising diffusion, and inverse folding models [18, 19] learn sequence–structure mappings from millions of examples. These structure-based methods are powerful when a target backbone is available but do not address the problem we consider: generating novel sequences from a family alignment alone.

All of these approaches, whether sequence-level or structure-conditioned, share a common requirement: large training corpora. Structure-conditioned models need libraries of backbone templates; sequence-level language models are typically pretrained on millions of sequences from databases such as UniRef50 and then fine-tuned or conditioned on family-specific alignments. This reliance on scale creates a fundamental blind spot. The median Pfam family contains fewer than 100 full-length members [20], and thousands of families have fewer than 50. The long tail of protein space extends further still: orphan families with no detectable homologs, newly discovered clades from metagenomic surveys, and engineered libraries from directed evolution campaigns may have only a handful of characterized sequences. In this scarce-data regime, learned generative models face a parameter-to-data ratio problem that manifests in predictable ways: VAEs and autoregressive models overfit to the training sequences and collapse in diversity, diffusion models fail to learn a meaningful denoising trajectory, and even pHMMs, despite their modest parameter count, emit sequences that drift far from the family’s functional envelope when the alignment is thin. Yet these small families are precisely where generative tools could have the greatest impact, by expanding sparse natural repertoires to enable functional characterization, guide directed evolution, or seed computational screens [21]. What is needed is an approach that can extract the statistical structure of a protein family from whatever data is available, even a few dozen sequences, without learning parameters that could overfit.

A seemingly unrelated line of work provides exactly this capability, by revealing that the attention mechanism at the heart of modern transformers is itself an energy-based model. Ramsauer et al. [22] showed that a single softmax attention operation is equivalent to one step of the retrieval dynamics on the modern Hopfield energy with a log-sum-exp interaction function, building on the dense

associative memory framework of Krotov and Hopfield [23]. This connection endows any set of stored examples with a well-defined energy landscape: the examples become memory patterns, the energy is minimized at (and near) those patterns, and the Boltzmann distribution over this landscape assigns high probability to states that resemble the stored set. Concurrently, score-based generative models [24, 25] and denoising diffusion models [26] have demonstrated that samples can be drawn from complex distributions by estimating the score function $\nabla \log p(\mathbf{x})$ and running Langevin dynamics. These methods achieve state-of-the-art sample quality in images, molecules, and protein structures, but they require a trained neural network to approximate the score. The Energy Transformer [27] exploits the Hopfield energy for discriminative tasks but does not sample from the associated Boltzmann distribution. The key insight we build on is that the Hopfield energy provides the score function *analytically*, as the residual of a single attention operation, which means that Langevin sampling can proceed without learning any parameters at all.

We propose a fundamentally different approach to protein sequence generation that requires no training. We observe that the modern Hopfield energy over a set of stored protein sequences defines a Boltzmann distribution whose score function is given in closed form by the residual of a single attention operation: $\nabla \log p_\beta(\xi) = \beta(\mathbf{T}(\xi) - \xi)$, where p_β is the Boltzmann density at inverse temperature β , ξ is the query state, and $\mathbf{T}$ is the softmax attention map. Because the score is exact, there is no score-estimation error, and the sampler inherits the convergence guarantees of the unadjusted Langevin algorithm (ULA) [28, 29]. Langevin dynamics on this energy yields a training-free sampler that we call *stochastic attention*: each iteration performs a softmax-weighted pull toward stored family members, a contraction toward the origin, and an isotropic Gaussian perturbation, requiring only two matrix-vector products per step ($\mathcal{O}(dK)$ cost, no GPU, no backpropagation). The inverse temperature β controls the balance between exploration and retrieval: at low β the sampler explores broadly, generating novel sequences that blend features from multiple stored patterns; at high β it retrieves sequences close to individual memories. Our prior work [30] introduced stochastic attention and validated it on synthetic data, MNIST digits, financial time series, and a single protein family (RRM). The present work extends this framework systematically to the protein domain with eight families, structure-level validation, a scaling analysis, and head-to-head comparison against protein-specific baselines.

The key question this paper addresses is: *what can you generate from a small protein family alignment with no training?* Across eight Pfam families spanning a ten-fold range of family sizes ($K=37\text{--}420$) and a seven-fold range of sequence lengths ($L=23\text{--}262$), we show that stochastic attention produces novel sequences that preserve family-level amino acid composition ($\text{KL} < 0.06$) and achieve substantial novelty (0.40–0.65 in PCA space), while folding into the correct three-dimensional architecture as confirmed by ESMFold [31] and AlphaFold2 [1] structure prediction with TM-align [32] structural comparison. We characterize the critical inverse temperature β^* at which the sampler transitions from diffuse exploration to pattern retrieval, and show that β^* can be predicted from PCA dimensionality alone ($R^2=0.97$), enabling fully automatic operation from a seed alignment. A systematic scaling study reveals stable generation quality down to $K=20$ sequences, the regime where learned models struggle most. Head-to-head comparison with profile HMMs [5], EvoDiff [14], and the MSA Transformer [13] reveals that stochastic attention uniquely balances compositional fidelity, sequence novelty, and structural viability without external training data, occupying a region of the quality–diversity–fidelity trade-off that no learned baseline achieves. Independent validation by ESM2-650M [31] confirms that generated sequences are recognized as protein-like by a language model trained on billions of evolutionary sequences, and cross-referencing against published deep mutational scanning data [33, 34] shows that over 90% of SA-generated substitutions correspond to experimentally tolerated mutations.

2 Results

We ran stochastic attention on all eight Pfam families and observed that the method consistently generated novel protein sequences while preserving family-level amino acid statistics, across a ten-fold range of family sizes and a seven-fold range of sequence lengths (Fig. 1; SI Appendix, Table S1).

We first examined whether the sampler converged to the correct Boltzmann distribution by comparing the outputs of the unadjusted Langevin algorithm (ULA) and its Metropolis-adjusted variant (MALA; see Materials and Methods). The MALA acceptance rates exceeded 99.6% in every family tested

(range: 99.6%–99.9%), confirming that the ULA discretization bias was negligible at step size $\alpha=0.01$. The two samplers produced near-identical metrics: for example, in the RRM family, SA retrieval achieved novelty 0.243 and KL divergence 0.107, while MALA yielded 0.249 and 0.112, respectively. This agreement held across all eight families, which justified the use of the computationally cheaper ULA in subsequent experiments.

We next assessed amino acid composition fidelity across families. In the generation regime (inverse temperature $\beta \approx 2\beta^*$, just above the entropy inflection point where pattern-specific basins first emerge; see Materials and Methods), stochastic attention (SA) achieved Kullback–Leibler (KL) divergences below 0.06 in every family, with several families below 0.02: WW achieved KL=0.008, Kunitz 0.013, and zinc finger 0.013. These values were substantially lower than the bootstrap baseline (KL range: 0.007–0.133), despite the fact that bootstrap samples are exact copies of stored patterns and thus trivially preserve composition at the individual-sequence level. The improvement arose because SA generation samples from the Boltzmann distribution over the entire family, producing a more representative ensemble than uniform resampling of a finite set. The convex combination baseline performed poorly, with KL divergences ranging from 0.06 (Pkinase) to 2.35 (zinc finger), which indicated that averaging across all stored patterns destroyed the peaked, position-specific residue distributions characteristic of conserved protein families (SI Appendix, Fig. S1).

We then examined whether the generated sequences were genuinely novel or merely replaying stored patterns. In the generation regime, novelty ranged from 0.40 (Defensin_beta) to 0.65 (WW), indicating substantial departure from any single stored memory (SI Appendix, Table S1). By contrast, bootstrap samples had zero novelty by construction, and Gaussian perturbation produced negligible novelty (< 0.02 in all families), confirming that small additive noise is insufficient to escape the basin of the nearest stored pattern. The retrieval regime produced intermediate novelty (0.17–0.35), consistent with samples that explore locally around stored patterns without fully escaping their basins. Nearest sequence identity in amino acid space confirmed this picture: generation-regime sequences had 52–66% identity to the nearest stored sequence, comparable to the typical within-family identity, while retrieval-regime sequences remained at 53–85% identity depending on the family’s intrinsic diversity.

We also characterized how the critical temperature varied across families. The empirical β^* ranged from 3.2 (PDZ, Pkinase) to 5.4 (WW), consistently lower than the $\sqrt{d}$ prediction for random patterns (where d is the principal component analysis [PCA] dimension retaining 95% of alignment variance) (5.8–13.6). The ratio $\beta^*/\sqrt{d}$ was approximately 0.4–0.6 across all families, which indicated that the structured similarity spectrum of real protein sequences, arising from conserved positions, correlated substitutions, and phylogenetic signal, shifts the retrieval-to-generation transition to lower temperatures. Families with higher mean per-column Shannon entropy ($\bar{H}_{\text{col}}$, measured in nats) tended to have lower $\beta^*/\sqrt{d}$ ratios, consistent with the intuition that more diverse families require less thermal energy to escape stored patterns. The effective number of sequences K_{eff} , computed by reweighting sequences at 80% identity to down-weight near-duplicates, equaled K in all but the WW family ($K_{\text{eff}}=419$ vs. $K=420$), indicating minimal redundancy in the Pfam seed alignments.

We next turned to the scaling study, in which we subsampled the WW domain at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition to characterize how generation quality, latent dimensionality, and the critical temperature co-vary with family size (SI Appendix, Fig. S2). Stochastic attention in the generation regime maintained low KL divergence across the entire range: KL=0.019 $\pm$ 0.008 at $K=20$, improving monotonically to KL=0.010 $\pm$ 0.002 at $K=400$. This graceful degradation contrasted sharply with the convex combination baseline, whose KL divergence exceeded 0.8 at every family size and worsened with increasing K (0.81 at $K=20$ to 1.50 at $K=400$), confirming that averaging over more patterns compounds the smoothing artifact. Bootstrap and Gaussian perturbation maintained moderate KL values (0.02–0.03) but produced no novelty at any K , making them ineffective as generative methods. The PCA dimensionality grew with family size, from $d=17$ –18 components at $K=20$ to $d=182$ –183 at $K=400$, and the critical temperature tracked this growth ($\beta^*=2.7$ to 5.5). Despite the six-fold increase in effective dimensionality, SA generation quality remained stable, which suggested that the entropy inflection criterion successfully adapted the operating temperature to the geometry of the memory matrix at each scale. Novelty increased with K (from 0.47 at $K=20$ to 0.74 at $K=400$), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56), indicating that the sampler explored further from stored patterns when more patterns were available to collectively shape the energy landscape.

We finally asked whether the critical temperature could be predicted from alignment statistics alone, without running an explicit temperature sweep (Fig. 2). We pooled 33 data points (the 8 Pfam families and 25 WW scaling replicates) and regressed β^* against MSA-derived features. We found that a simple two-parameter model, $\beta^* \approx 1.57 + 0.28\sqrt{d}$ (intercept 1.57 ± 0.07 , slope 0.281 ± 0.007 ; $\pm$ denotes bootstrap standard error [SE], 10,000 resamples), explained 97% of the variance ($R^2=0.97$, root-mean-square error [RMSE]=0.16), with d the PCA dimension at 95% retained variance. Adding further predictors (mean column entropy $\bar{H}_{\text{col}}$, effective number of sequences K_{eff} , or spectral concentration $\lambda_1/\text{tr}(\mathbf{C})$, the fraction of total variance explained by the leading principal component) produced negligible improvement ($\Delta R^2 < 0.001$), which indicated that PCA dimensionality captured most of the relevant variation. The fitted slope of 0.28 was roughly half the $\sqrt{d}$ coefficient predicted for random patterns, consistent with the empirical ratio $\beta^*/\sqrt{d} \approx 0.4\text{--}0.6$ observed across families. Importantly, the naive random-pattern prediction $\beta^* = \sqrt{d}$ yielded RMSE=5.3, a $33\times$ larger error, confirming that the structured similarity spectrum of real protein families shifts the phase transition to substantially lower temperatures. This regression provides a practical shortcut: given a new family, one can estimate β^* from PCA alone and set the generation temperature without a temperature sweep.

We compared stochastic attention against three established learned baselines: profile HMM emit, EvoDiff (MSA-conditioned diffusion), and the MSA Transformer (Gibbs masked language model sampling) across all eight families (Fig. 3). We used Wilcoxon rank-sum tests on per-chain estimates (30 chains $\times$ 5 samples) with Cohen’s d effect sizes. EvoDiff achieved the lowest KL divergence in the seven shorter families (0.003–0.011), consistent with its pretraining on millions of UniRef50 MSAs, and profile HMM emit also produced low KL values (0.006–0.031) by directly capturing position-specific residue frequencies. Stochastic attention generation was competitive with both (0.008–0.060) despite using only the family’s seed alignment, typically 37–420 sequences, with no external training data. However, EvoDiff’s performance degraded sharply on the longest family: for Pkinase ($L=262$), EvoDiff produced a KL divergence of 0.20, an order of magnitude worse than SA (0.04) or HMM emit (0.02), indicating that the discrete diffusion process struggles to maintain compositional fidelity over long sequences. Generating 150 Pkinase sequences with EvoDiff required ~ 14 hours on CPU, compared to seconds for SA. However, the three metrics together revealed a trade-off that no single learned baseline resolved. The critical metric was nearest sequence identity, where all three baselines differed significantly from SA generation ($p < 0.001$ in all families). Profile HMM emit produced the highest novelty (0.56–0.71) but the lowest sequence identity to stored patterns (11–42%; SA gen vs. HMM emit: $d = 3.6\text{--}24.6$ across families), indicating sequences so divergent from the family that they likely fall outside its functional envelope. The MSA Transformer showed a similar pattern: high novelty (0.32–0.67) but low sequence identity (9–59%; $d = 1.1\text{--}26.8$). EvoDiff produced intermediate sequence identity (0.15–0.57) but still significantly lower than SA generation in every family ($p < 0.001$, $d = 1.6\text{--}16.3$). Stochastic attention generation uniquely occupied the regime of low KL divergence, substantial novelty (0.40–0.65), and moderate sequence identity (52–66%), generating sequences that departed meaningfully from stored patterns while remaining within the family’s characteristic sequence space. Critically, SA achieved this balance without pretraining, backpropagation, or GPU resources, using only the information present in the seed alignment itself. As a further comparison, we fitted a Potts model via pseudo-likelihood maximization (plmDCA) to the SH3 and Kunitz alignments and generated sequences by Gibbs sampling (SI Appendix, Table S4). The Potts model, which captures pairwise coevolutionary couplings with approximately 500,000 fitted parameters, achieved comparable composition fidelity and novelty to SA in both families, but with slightly lower sequence identity to stored patterns (0.54 vs. 0.59 for SH3; 0.55 vs. 0.61 for Kunitz). The similar output quality suggests that the information captured by pairwise couplings is also implicitly encoded in the Hopfield energy landscape; the critical difference is that SA requires no parameter fitting and runs in seconds rather than minutes.

The beta defensin family (PF00711) provided an opportunity to test whether SA-generated sequences preserved the biophysical properties required for antimicrobial function, because beta defensins have well-characterized compositional requirements: a net positive charge at physiological pH, a moderate hydrophobic ratio, and a conserved cysteine scaffold that forms three disulfide bonds essential for the beta-defensin fold. We computed net charge (summing +1 for Arg and Lys, -1 for Asp and Glu), hydrophobic residue fraction, and cysteine count for all generated and stored sequences (see Materials and Methods). SA-generated defensin sequences preserved the six-cysteine motif with near-perfect fidelity (6.01 ± 0.08 mean cysteines, compared to 6.02 ± 0.15 for natural sequences), while simultaneously producing the highest mean net charge ($+6.1 \pm 1.6$) and the tightest charge

distribution of any method. When scored against a composite antimicrobial plausibility filter (net charge $\geq +2$, hydrophobic ratio in $[0.3, 0.7]$, ≥ 4 cysteines), 51% of SA-generated sequences passed, compared to 42% of natural stored sequences, 53% for EvoDiff, 25% for profile HMM emit, and 16% for the MSA Transformer. SA generation thus matched or exceeded the antimicrobial plausibility of natural defensins without any explicit biophysical objective, relying only on the compositional constraints implicitly encoded in the family alignment.

We assessed structural plausibility by predicting folded structures for 50 sequences per method per family using ESMFold (Fig. 4), comparing SA generation to stored (natural) sequences via Wilcoxon rank-sum tests with Cohen’s d effect sizes. We used two complementary structure metrics: the predicted local distance difference test (pLDDT), a per-residue folding confidence score where values above 70 indicate a confident prediction, and the template modeling score (TM-score), a global measure of structural similarity where values above 0.5 indicate the same fold. SA-generated sequences achieved mean pLDDT scores statistically indistinguishable from stored natural sequences in five of seven families (Wilcoxon $p > 0.05$; RRM: 76.2 vs. 76.9, $p=0.45$; WW: 83.7 vs. 82.9, $p=0.08$; zf-C2H2: 83.4 vs. 81.2, $p=0.23$; PDZ: 77.6 vs. 77.4, $p=0.55$; Pkinase: 88.7 vs. 89.7, $p=0.11$). In Kunitz, SA generation achieved significantly *higher* pLDDT than stored sequences (90.8 vs. 89.3, $p=0.001$, $d=0.67$). Only SH3 showed lower pLDDT for SA generation (70.5 vs. 74.1, $p=0.006$, $d=-0.56$), though the mean remained above the pLDDT=70 confidence threshold; as described below, this deficit was not confirmed by AlphaFold2 cross-validation, suggesting it reflects a predictor-specific bias rather than a genuine structural deficiency. Across all families, $\geq 86\%$ of SA-generated sequences exceeded pLDDT=70 (with 100% in WW, Kunitz, and Pkinase). TM-scores to family reference structures provided an even stronger result. In five of seven families, SA generation achieved significantly *higher* TM-scores than stored sequences (Wilcoxon $p < 0.001$; RRM: 0.409 vs. 0.380, $d=1.62$; WW: 0.181 vs. 0.174, $d=1.14$; Kunitz: 0.847 vs. 0.828, $d=0.99$; PDZ: 0.638 vs. 0.574, $d=1.41$; Pkinase: 0.707 vs. 0.679, $d=1.56$). The remaining two families showed no significant difference (SH3: $p=0.56$; zf-C2H2: $p=0.11$). The large effect sizes ($d > 1.0$) in several families indicate that this is not merely a failure to reject the null: SA-generated sequences are structurally *more similar* to the canonical family fold than a random sample of natural family members. This likely reflects the Boltzmann ensemble’s tendency to weight the central tendency of the family distribution, producing sequences closer to the “consensus fold” than individual natural sequences, which may contain lineage-specific structural deviations. The three families with uniformly low TM-scores (RRM, WW, zf-C2H2) showed the same pattern for stored sequences, indicating a length mismatch between the generated domain and the multi-domain reference structure rather than a failure of the generative method. As an independent cross-validation, we repeated the structure prediction for all families with completed AlphaFold2 runs (SH3, WW, Kunitz, zf-C2H2; additional families are in progress) and performed the same statistical comparisons (Fig. 4). The AF2 results corroborated the ESMFold findings while resolving the single negative result. For SH3, the only family where ESMFold had indicated a significant pLDDT deficit for SA generation, AlphaFold2 showed no difference (SA: 90.7 vs. Stored: 89.3, $p=0.99$) and revealed significantly higher TM-scores for SA-generated sequences (0.736 vs. 0.721, $p < 0.001$, $d=0.72$), with a TM-score well above the 0.5 same-fold threshold. The WW and zf-C2H2 families showed the same pattern as ESMFold: SA-generated sequences matched or exceeded stored sequences in pLDDT (WW: 84.7 vs. 83.1, $p=0.005$; zf-C2H2: 92.8 vs. 89.2, $p < 0.001$) and TM-score (WW: 0.182 vs. 0.172, $p < 0.001$, $d=1.73$; zf-C2H2: 0.254 vs. 0.242, $p=0.036$, $d=0.47$). Kunitz showed the same pattern: pLDDT was statistically indistinguishable (SA: 94.9 vs. Stored: 93.3, $p=0.18$) while TM-scores were significantly higher for SA generation (0.850 vs. 0.830, $p < 0.001$, $d=0.73$), consistent with the strong ESMFold results for this family and representing the highest absolute TM-score of any family tested. The concordance between the two predictors across four families, combined with the resolution of the SH3 anomaly, strengthens the conclusion that SA-generated sequences fold correctly across all families tested. To complement the aggregate statistics, we selected the highest-confidence SA-generated structure from each family and superimposed it on the experimentally determined reference structure (Fig. 5). The structural overlays confirm, at the level of individual folds, that SA-generated sequences adopt the canonical architecture of their target family with sub-angstrom backbone RMSD, despite having no sequence identity to the reference and 40–65% novelty relative to all stored family members.

As a complementary, sequence-level assessment of biological plausibility, we scored all SA-generated and stored sequences using ESM2-650M [31], a protein language model pretrained on millions of UniRef50 sequences that is entirely independent of the stochastic attention framework (SI Appendix,

Table S2). We computed pseudo-perplexity via masked marginal scoring, where lower values indicate a more protein-like sequence. SA-generated sequences were statistically indistinguishable from natural family members in five of eight families (Wilcoxon $p > 0.05$), with SA achieving significantly better scores in the Kunitz domain (4.47 vs. 4.98, $p < 0.001$, $d = -0.74$). Three families showed modestly higher pseudo-perplexity for SA generation (SH3, WW, PDZ; $d = 0.6-1.0$), though all SA-generated sequences remained well within the range of natural protein sequences (overall mean: 5.71 vs. 5.58). This independent validation confirms that SA-generated sequences are not merely structurally plausible but are recognized as protein-like by a language model trained on billions of evolutionary sequences. We further cross-referenced SA-generated substitutions against published deep mutational scanning (DMS) datasets for the PDZ domain (McLaughlin et al. [33]) and the SH3 domain (Faure et al. [34]), finding that 91.7% and 90.4% of matched substitutions, respectively, corresponded to experimentally tolerated mutations, significantly exceeding the baseline tolerance rates of 77.5% and 67.5% in the DMS datasets (binomial $p < 10^{-135}$ and $p < 10^{-156}$; SI Appendix, Table S3). The continuous DMS fitness scores of SA-generated substitutions were also significantly higher than those of random single mutations (Cohen’s $d = 0.42$ and 0.70), confirming that the Boltzmann sampling procedure preferentially generates substitutions that have been experimentally validated as functional.

To understand how SA-generated sequences achieve structural fidelity while maintaining high novelty, we examined the position-specific substitution pattern of the Kunitz domain in detail (Fig. 6). We aligned the five highest-confidence SA-generated sequences (AlphaFold2 pLDDT ≥ 96) against the family consensus and colored each position by its conservation level in the stored MSA. The pattern was striking: all five sequences preserved every highly conserved position (11 of 11 positions with $> 90\%$ conservation) and all six cysteines that form the three disulfide bonds defining the Kunitz fold, while introducing 17–25 amino acid substitutions concentrated at evolutionarily variable positions. No substitutions occurred at the structural glycines (positions 9, 34, 37) that enable tight turns, or at the phenylalanine and asparagine residues (positions 30, 38, 40, 42) that anchor the hydrophobic core. The per-position Shannon entropy of SA-generated sequences correlated almost perfectly with that of the stored MSA ($r = 0.968$), confirming that the Boltzmann sampling procedure does not merely preserve global amino acid frequencies but faithfully recapitulates the position-specific conservation landscape of the family. Positions that are constrained by structure or function in natural sequences remain invariant in the generated ensemble, while positions that tolerate variation in nature exhibit comparable diversity in the generated output. This selectivity arises without any explicit conservation mask or position-specific prior: it emerges naturally from the energy landscape defined by the stored patterns, in which conserved positions create deep, narrow wells that the sampler cannot escape, while variable positions define broad, shallow basins that permit exploration.

The preceding analysis established that SA preserves the single-site conservation landscape of each family. We next asked whether this fidelity extends to pairwise residue covariation, which reflects structural contacts and functional couplings that single-site statistics cannot capture. We computed pairwise mutual information (MI) matrices for the stored and SA-generated alignments in each family and correlated the $L(L-1)/2$ upper-triangle MI values (SI Appendix, Fig. S3, Table S5). Across seven of the eight families, the Pearson correlation between stored and SA-generated MI vectors ranged from $r = 0.53$ to 0.92 (mean $r = 0.72$ excluding Pkinase), confirming that SA-generated sequences preserve the pairwise coupling landscape of their parent family. The strongest MI preservation occurred in families with large K relative to L (zf-C2H2: $r = 0.92$; Defensin_beta: $r = 0.85$), while the lowest occurred in Pkinase ($r = 0.05$), where only 37 sequences span 262 alignment positions, a regime in which MI estimation is unreliable for any method. Notably, SA generation substantially outperformed profile HMMs in every family (e.g., Kunitz: $r = 0.80$ vs. 0.32 ; WW: $r = 0.71$ vs. 0.29), which confirmed that the Hopfield energy captures correlational structure beyond what position-independent models encode. As expected, methods that explicitly model pairwise couplings (Potts/plmDCA: $r = 0.75-0.89$ in SH3 and Kunitz) or that leverage millions of pretraining MSAs (EvoDiff: $r = 0.59-0.97$) achieved higher MI correlations than SA, which fits no parameters and uses only the seed alignment. For three families where the reference PDB structure closely matched the alignment length (Kunitz, SH3, Defensin_beta), we mapped structural contacts ($C\alpha-C\alpha$ distance $< 8 \text{ \AA}$, sequence separation ≥ 5) to alignment positions and found that MI at contact pairs in the SA-generated ensemble tracked MI at those same pairs in the stored alignment ($r = 0.49-0.86$; SI Appendix, Fig. S3), confirming that covariation preservation extends to structurally relevant positions.

3 Discussion

We set out to ask what can be generated from a small protein family alignment with no training, and the answer is surprisingly rich: sequences that are novel, compositionally faithful, and structurally correct, produced in seconds on a laptop from nothing but the alignment itself. Across eight Pfam families spanning a ten-fold range of family sizes ($K=37\text{--}420$) and a seven-fold range of sequence lengths ($L=23\text{--}262$), stochastic attention matched or exceeded the output quality of methods pretrained on millions of sequences, while requiring zero external data, zero training, and zero GPU resources. The generated sequences preserve family-level amino acid composition ($KL < 0.06$), achieve substantial novelty ($0.40\text{--}0.65$), and fold into the correct three-dimensional architecture as confirmed by both ESMFold and AlphaFold2 structure prediction. This combination of properties is not achieved by any single learned baseline we tested: profile HMMs and the MSA Transformer generate sequences that drift outside the family, EvoDiff requires millions of pretraining sequences and hours of compute, and all three sacrifice sequence identity to stored patterns in ways that stochastic attention does not. The fact that a training-free method, one that treats the seed alignment as both the model and the data, can compete with systems built on orders of magnitude more information suggests that the statistical structure of protein families is rich enough to support generation without learned representations, provided the generative framework respects the geometry of the family’s sequence space.

The structure validation results were particularly striking. Rather than merely matching the structural quality of natural sequences, SA-generated sequences achieved significantly higher TM-scores to the canonical family fold in five of seven families by ESMFold (Wilcoxon $p < 0.001$, Cohen’s $d > 1.0$). This was not an artifact of sequence similarity: the same generated sequences had 40–65% novelty relative to stored patterns. Two (non-exclusive) explanations are possible. First, the Boltzmann ensemble defined by the modern Hopfield energy weights the central tendency of the family distribution, producing sequences that are closer to a “consensus fold” than individual natural sequences, which carry lineage-specific structural deviations accumulated over evolutionary time. Second, the structure predictors themselves (ESMFold, AlphaFold2) may be biased toward consensus-like sequences: models trained on evolutionary data could assign higher confidence to sequences that resemble a family’s central tendency, inflating TM-scores for consensus-proximal inputs regardless of true structural quality. The fact that both predictors agree, and that the effect persists across families with different levels of structural diversity, favors the consensus-fold explanation, but a definitive resolution would require experimental structure determination of selected generated sequences. The pLDDT confidence scores told a complementary story: SA-generated sequences were statistically indistinguishable from stored sequences in five of seven families by ESMFold ($p > 0.05$), confirming that the sampler does not sacrifice folding confidence for diversity. The single exception, SH3, where ESMFold reported a modest pLDDT deficit (70.5 vs. 74.1), was resolved by AlphaFold2 cross-validation, which showed no pLDDT difference (90.7 vs. 89.3, $p=0.99$) and significantly higher TM-scores for SA generation (0.736 vs. 0.721, $p < 0.001$). The concordance between two independent structure predictors, and the resolution of the only negative result as a predictor-specific artifact, addresses the most immediate reviewer concern for any sequence-level generative method (do the sequences actually fold?) with a clear affirmative.

Comparison against three established baselines (profile HMM emit, EvoDiff, and the MSA Transformer) revealed a trade-off that stochastic attention uniquely resolves. All three baselines produced sequences with significantly lower identity to stored family members than SA generation ($p < 0.001$ in every family, with effect sizes $d = 1.1\text{--}26.8$). Profile HMM emit and the MSA Transformer generated sequences at 9–59% identity, well below the typical within-family range, suggesting sequences that have drifted outside the family’s functional envelope. EvoDiff balanced the metrics more effectively in shorter families but required pretraining on millions of UniRef50 MSAs, scaled poorly with sequence length (~14 hours for 150 Pkinase sequences on CPU, compared to seconds for SA), and suffered a sharp degradation in compositional fidelity on the longest family (Pkinase $KL=0.20$, an order of magnitude worse than SA). Stochastic attention occupied the middle ground (competitive KL divergence, high novelty, moderate sequence identity) using only the seed alignment. A targeted comparison with a Potts model fitted via pseudo-likelihood maximization (plmDCA) on two families (SI Appendix, Table S4) revealed that pairwise coevolutionary models achieve comparable output quality to SA, despite fitting approximately 500,000 parameters to fewer than 100 sequences. This convergence is informative: it suggests that the modern Hopfield energy, through the softmax attention operation, implicitly captures the same correlational structure that Potts models fit explicitly, but without parameter estimation. The pairwise mutual information analysis (SI

Appendix, Table S5) provides direct evidence for this implicit coupling: SA-generated sequences preserved the pairwise MI landscape of the stored alignment (Pearson $r=0.53$ – 0.92 across seven families), consistently exceeding the MI preservation of profile HMMs ($r=0.07$ – 0.83), which model each position independently. That SA captures substantial pairwise structure without fitting any parameters is consistent with the exponential storage capacity of modern Hopfield networks, in which the softmax attention operation couples all positions simultaneously through the stored-pattern energy landscape. Methods that fit pairwise couplings explicitly (Potts) or draw on large-scale pretraining (EvoDiff) achieve stronger MI preservation, reflecting their direct access to pairwise or higher-order statistics; the relevant comparison for SA, however, is against single-site models, where the improvement demonstrates that the Hopfield energy encodes information beyond marginal frequencies. The practical advantage is that SA avoids the fitting step entirely, which becomes increasingly fragile as the data-to-parameter ratio falls below 10^{-4} , a regime that encompasses most Pfam families. This makes the method particularly attractive for orphan families, newly discovered protein clades, or engineered libraries where external training corpora are unavailable or inappropriate.

The beta defensin results illustrate a broader principle: the Boltzmann ensemble over a family alignment implicitly enforces the compositional constraints required for function, without any explicit biophysical objective. SA-generated defensin sequences preserved the cysteine scaffold, maintained the positive net charge characteristic of antimicrobial peptides, and matched or exceeded the antimicrobial plausibility of natural defensins on a composite biophysical filter (Results). This implicit enforcement arises because the energy landscape concentrates probability near the stored patterns, which already satisfy the biophysical constraints by virtue of being functional proteins. The result is especially relevant given the growing need for novel antimicrobial peptides in the face of rising antibiotic resistance [35], and suggests that stochastic attention could serve as a rapid, training-free tool for expanding natural AMP repertoires.

The scaling study on the WW domain showed that this balance is not fragile: stochastic attention maintained stable generation quality down to $K=20$ sequences, with KL divergence below 0.02 and novelty above 0.47 even at this extreme. This robustness arises from a structural property of the method: the energy landscape is defined entirely by the stored patterns, with no parameters that could overfit to a small training set. There is no encoder to memorize, no decoder to collapse, and no denoising network to undertrain. The entropy inflection criterion automatically adapts the operating temperature to the geometry of the memory matrix at each scale, and the critical temperature β^* itself can be predicted from PCA dimensionality alone ($R^2=0.97$), eliminating the need for a temperature sweep and enabling fully automatic operation from a seed alignment. In contrast, learned generative models face a fundamental parameter-to-data ratio problem when $K < 100$: VAEs and diffusion models have thousands to millions of parameters that cannot be meaningfully constrained by a few dozen sequences, and even the most parameter-efficient approaches (e.g., profile HMMs) emit sequences that lack the higher-order correlations needed for structural integrity. This regime, $K < 100$, encompasses the majority of Pfam families and the entirety of the long tail of protein space where generative tools are most needed.

Several limitations should be acknowledged. The most significant is that stochastic attention operates on a fixed alignment representation: it requires a pre-computed MSA and a fixed alignment length L , and cannot generate insertions, deletions, or variable-length sequences. This constraint is inherited from the one-hot-plus-PCA encoding, which maps each sequence to a vector of fixed dimension $21L$; any change in L would alter the representation space itself. In practice, this means that the method is best suited to compact, well-aligned domains rather than multi-domain proteins or families with frequent large insertions. Extending stochastic attention to variable-length generation would likely require replacing PCA with a length-agnostic embedding (e.g., learned or kernel-based) while preserving the closed-form score function, a nontrivial but tractable direction for future work. The PCA projection discards information that may be relevant for capturing higher-order correlations, and the choice of retained variance (95%) involves a trade-off between compactness and expressiveness that we have not optimized. The scaling study used only WW domain subsamples; although the eight-family cross-family analysis provides breadth, a multi-family scaling study would strengthen claims about general robustness to small K . The multi-chain initialization strategy, while effective for the families studied here, does not guarantee coverage of all modes in the Boltzmann distribution, particularly for very large or highly diverse families. The eight families studied here are well-characterized domains with abundant structural and functional annotation, which facilitated validation but may not represent the full diversity of protein space; performance on disordered

regions, transmembrane domains, or families with very low sequence identity remains untested. Finally, our structure validation used ESMFold and AlphaFold2, which are themselves neural models with biases; wet-lab characterization of selected generated sequences would provide stronger evidence of functional viability and is a natural next step.

Looking beyond proteins, the connection between attention and Boltzmann sampling suggests a broader principle: any set of examples, regardless of domain, defines an energy landscape via the modern Hopfield energy, and Langevin dynamics on this landscape provides a principled generative model with no training. The requirements are minimal: the examples must be representable as vectors in a common space, and the practitioner must choose an inverse temperature that balances fidelity and diversity. This perspective may extend naturally to other biological sequence families where data is scarce but examples are structured, such as antibody repertoires (where individual clonotypes may have only a handful of somatic variants), RNA families (where Rfam seed alignments are typically smaller than their Pfam counterparts), or small-molecule libraries (where chemical series may contain only tens of analogs). Beyond biology, any domain with a small, curated set of high-quality examples, such as materials compositions, musical motifs, or architectural floor plans, could in principle benefit from the same framework. The analytic score function also provides a natural interface with the convergence theory of Langevin dynamics [28, 29], enabling formal guarantees on sample quality that are absent from most neural generative models, and opening a path toward certifiable generation in safety-critical applications such as therapeutic protein design.

In summary, stochastic attention is a competitive, training-free alternative to learned generative models for protein sequence generation, particularly in the small-family regime that characterizes most of protein space. The method requires only a seed alignment, has $\mathcal{O}(dK)$ per-step cost, needs no GPU, and produces structurally plausible sequences whose diversity and composition match, or exceed, those of natural family members.

4 Materials and Methods

4.1 From Hopfield Energy to Score Function

Let $\mathbf{X} = [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ be the memory matrix whose columns are vectorized protein sequences from a family alignment of K members, each represented in d dimensions (after encoding and, optionally, PCA projection). The modern Hopfield energy [22] is

$$E(\boldsymbol{\xi}) = \frac{1}{2} \|\boldsymbol{\xi}\|^2 - \frac{1}{\beta} \log \sum_{k=1}^K \exp(\beta \mathbf{m}_k^\top \boldsymbol{\xi}), \quad (1)$$

where $\beta > 0$ is an inverse temperature. The gradient is $\nabla E(\boldsymbol{\xi}) = \boldsymbol{\xi} - \mathbf{T}(\boldsymbol{\xi})$, where

$$\mathbf{T}(\boldsymbol{\xi}) := \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi}) \quad (2)$$

is the softmax attention retrieval map [22]. The Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$ has score function

$$\nabla \log p_\beta(\boldsymbol{\xi}) = -\beta \nabla E(\boldsymbol{\xi}) = \beta (\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi}). \quad (3)$$

This score is *exact* and computed by a single attention operation; no score network or training is required.

Applying the unadjusted Langevin algorithm (ULA) [36] to the potential $U = \beta E$ with step size $\tilde{\alpha} = \alpha/\beta$ yields the *stochastic attention* update:

$$\boldsymbol{\xi}_{t+1} = (1 - \alpha) \boldsymbol{\xi}_t + \alpha \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi}_t) + \sqrt{\frac{2\alpha}{\beta}} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (4)$$

Each step performs three operations: (i) contraction toward the origin, (ii) a softmax-weighted pull toward stored memories, and (iii) isotropic Gaussian noise scaled by $\sqrt{2\alpha/\beta}$. The per-step cost is $\mathcal{O}(dK)$: two matrix-vector products and one softmax. As $\beta \rightarrow \infty$ the noise vanishes and the update reduces to deterministic retrieval; as $\beta \rightarrow 0$ sampling becomes noise-dominated.

4.2 Critical Temperature from PCA Dimensionality

The attention entropy $H(\beta) = -\sum_k p_k \log p_k$, where $p_k = \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi})_k$, quantifies selectivity: $H = \log K$ when attention is uniform ($\beta \rightarrow 0$) and $H \rightarrow 0$ when attention concentrates on a single memory ($\beta \rightarrow \infty$). The entropy derivative satisfies [30]

$$\frac{dH}{d\beta} = -\beta \text{Var}_{\mathbf{p}}(e), \quad (5)$$

where $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ are the query-memory similarities. The maximum rate of entropy loss occurs at the inflection point β^* satisfying $d^2 H/d\beta^2 = 0$, which marks the onset of pattern-specific structure in the energy landscape. The form of β^* is constrained by the geometry of the stored patterns. Because the memory columns lie on the unit sphere $\mathbb{S}^{d-1}$, concentration of measure dictates that the similarity variance is $\text{Var}(e_k) = 1/d$ exactly for any query drawn uniformly from $\mathbb{S}^{d-1}$ (Appendix S9). The characteristic similarity scale is therefore $\sigma = 1/\sqrt{d}$, and since β enters the softmax as βe_k , the inflection must occur when $\beta\sigma = \mathcal{O}(1)$, giving $\beta^* \sim \sqrt{d}$. A separate calculation shows that the softmax entropy over K i.i.d. Gaussian scores undergoes its inflection at a dimensionless temperature $\tau^* \approx 1.6$ that is nearly independent of K for $K \geq 30$ (Appendix S9), providing a universal baseline for the transition.

Empirically, we find a simple linear relationship across eight Pfam families and WW-domain subsamples spanning $K = 20$ –420 and $d = 18$ –186 (Results):

$$\beta^* \approx 1.57 + 0.28\sqrt{d}, \quad (6)$$

which captures 97% of the variance ($R^2 = 0.97$; OLS $R^2 = 0.962$, bootstrap median $R^2 = 0.969$). The intercept $1.57 \approx \tau^*$ recovers the universal softmax inflection baseline, confirming that the same dimensionless transition governs both idealized Gaussian scores and real protein similarities. The coefficient 0.28 of $\sqrt{d} = 1/\sigma$ is substantially smaller than the pure Gaussian prediction $\tau^*/\sigma \approx 1.6\sqrt{d}$; this reduction arises because β^* is computed at stored patterns $\mathbf{m}_k$, where the self-similarity $\mathbf{m}_k^\top \mathbf{m}_k = 1$ anchors one score at the maximum of $\mathbb{S}^{d-1}$ and makes concentration easier than for a random query (Appendix S9). From a practical standpoint, Eq. (6) enables fully automatic operation: given a seed alignment, d is determined by PCA at 95% variance, and β^* follows without a temperature sweep.

4.3 Regularity and Convergence

The modern Hopfield energy has Lipschitz-continuous gradient with constant $L = 1 + \beta\sigma_{\max}^2/2$, where $\sigma_{\max} = \|\mathbf{X}\|_{\text{op}}$, and satisfies the dissipativity condition $\langle \nabla E(\boldsymbol{\xi}), \boldsymbol{\xi} \rangle \geq \|\boldsymbol{\xi}\|^2/2 - M^2/2$ with $M = \max_k \|\mathbf{m}_k\|$ [30]. When $\beta\sigma_{\max}^2 < 2$, the energy is strictly convex and ULA iterates converge to p_β in Wasserstein-2 distance at a geometric rate [29, 28].

In the protein setting, operating temperatures of interest satisfy $\beta\sigma_{\max}^2 \gg 2$, placing the sampler in a non-convex regime where the energy landscape has multiple metastable basins concentrated near the stored patterns. Following standard practice for multimodal Langevin sampling [37], we adopt a multi-chain protocol: N_c independent chains are initialized near distinct stored patterns, exploiting fast intra-basin mixing while achieving inter-basin coverage through diverse initialization. The Metropolis-adjusted variant (MALA) provides an acceptance-rate diagnostic: acceptance rates near 1.0 confirm that ULA discretization bias is negligible for the chosen step size α .

We applied stochastic attention to protein sequence generation across eight Pfam families and conducted a systematic scaling study to characterize performance as a function of family size. All experiments used the unadjusted Langevin algorithm (ULA) with step size $\alpha = 0.01$. For each family and condition, we ran 30 independent chains of $T = 5,000$ iterations, initialized near randomly chosen stored patterns with Gaussian perturbation ($\sigma_{\text{init}} = 0.01$). After discarding 2,000 burn-in iterations and thinning every 100 steps, we retained 5 evenly spaced samples per chain for a total of $S = 150$ generated sequences per condition. Metropolis-adjusted Langevin algorithm (MALA) chains were run in parallel with identical parameters to provide acceptance-rate diagnostics. Throughout, we used PCA dimensionality reduction retaining 95% of the variance in the one-hot encoded alignment, followed by unit-norm projection to form the memory matrix $\tilde{\mathbf{X}}$.

The inverse temperature was set automatically for each family via the entropy inflection criterion described above. We identified β^* from the attention entropy curve and operated in two regimes,

both above β^* . Below β^* the energy landscape is essentially flat and Langevin dynamics degenerates into a random walk on the sphere, producing sequences with no family-specific structure. Above β^* , metastable basins form around stored patterns and the sampler begins to produce protein-like sequences. We therefore defined a *generation* regime at $\beta_{\text{gen}} \approx 2\beta^*$, just above the transition, where the basins are shallow enough that the sampler can hop between them, producing novel recombinations of family features. At much higher inverse temperature, we defined a *retrieval* regime at $\beta_{\text{ret}} \approx 20\beta^*$, where the basins are deep and each chain remains trapped near its initialization pattern, producing near-copies of stored sequences. Three simple baselines were evaluated under identical conditions: bootstrap replay (uniform resampling of stored patterns), Gaussian perturbation (stored pattern plus isotropic noise matched to the SA noise scale), and random convex combination (Dirichlet-weighted average of all stored patterns).

We evaluated generated sequences using five metrics. *Novelty* measured departure from stored patterns as $1 - \max_k \cos(\hat{\mathbf{x}}, \mathbf{m}_k)$ in PCA space. *Diversity* was the mean pairwise cosine distance among generated samples. *Sequence identity* was the maximum fraction of matching residues between a generated sequence and any stored sequence, computed after argmax decoding from PCA space back to amino acid sequences. *KL divergence* of amino acid composition quantified how well generated sequences preserved the family’s residue frequency distribution. *Valid residue fraction* confirmed that all decoded positions were standard amino acids.

Protein families and scaling study. We selected eight families spanning a range of sizes, sequence lengths, and conservation levels (Table 1): RRM (PF00076, $K=68$, $L=71$), SH3 (PF00018, $K=55$, $L=48$), WW (PF00397, $K=420$, $L=31$), Kunitz (PF00014, $K=99$, $L=53$), zinc finger C2H2 (PF00096, $K=151$, $L=23$), PDZ (PF00595, $K=44$, $L=83$), protein kinase (PF00069, $K=37$, $L=262$), and beta defensin (PF00711, $K=45$, $L=36$). Seed alignments were downloaded from InterPro [20] and cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps. The resulting PCA dimensions ranged from $d=34$ (Pkinase) to $d=186$ (WW). To characterize how generation quality degrades with decreasing family size, we also conducted a scaling study in which we fixed the WW domain (the largest family, $K=420$) and subsampled to $K \in \{20, 50, 100, 200, 400\}$. At each value of K we drew five independent random subsets, rebuilt the memory matrix from scratch (re-computing PCA and β^*), ran stochastic attention and all baselines, and recorded metrics; this design isolates the effect of family size from family identity, and the five repeats provide error estimates.

Learned baselines. To contextualize SA against methods that require pretraining on large external corpora, we compared three established generative approaches for protein sequences. (i) *Profile HMM emit*: we built a profile HMM from each family’s seed alignment using HMMER3 [5] (`hmmbuild -amino`) and emitted $S=150$ sequences with `hmmemit -N 150 -seed 42`. Emitted sequences were filtered to standard amino acids and truncated or padded to the alignment length L . (ii) *EvoDiff*: we used the MSA-conditioned order-agnostic discrete diffusion model (MSA_OA_DM_RANDSUB) [14], which was pretrained on UniRef50 MSAs. Generation used the default denoising schedule with MaxHamming subsampling of the conditioning MSA (depth ≤ 64). (iii) *MSA Transformer*: we used the 100M-parameter MSA Transformer [13] (`esm_msa1b_t12_100M_UR50S`) for generation via iterative Gibbs-like masked language model sampling: a fully masked query row was appended to the family MSA (depth ≤ 128), and 50 rounds of masking 15% of positions and resampling from the softmax were performed. For each baseline, the same 150-sequence budget and the same evaluation metrics (KL, novelty, sequence identity) were applied. EvoDiff was run on CPU; for the Pkinase family ($L=262$), a single sequence required ~ 7 minutes, yielding ~ 14 hours for 150 sequences (compared to seconds for SA).

Structure validation. To assess whether generated sequences adopt structures consistent with their target family, we predicted the three-dimensional structure of 50 sequences per method per family using the ESMFold API [31]. For each predicted structure, we extracted the mean predicted local distance difference test (pLDDT) score from the B-factor column as a measure of folding confidence (pLDDT > 70 indicates a confident prediction). We then compared each predicted structure to a representative experimentally determined structure for the family using TM-align [32] (TM-score > 0.5 indicates the same fold). Reference structures were obtained from the Research Collaboratory for Structural Bioinformatics Protein Data Bank (RCSB PDB): 1FXL (RRM), 1SHG (SH3), 1PIN (WW), 1BPI (Kunitz), 1ZAA (zf-C2H2), 1BE9 (PDZ), 1ATP (Pkinase), and 1E4S (Defensin_beta).

As a positive control, we also predicted structures for 50 stored (natural) sequences from each family, providing an upper bound on expected pLDDT and TM-score. As an independent cross-validation, we repeated the structure prediction for all eight families using AlphaFold2 [1] via ColabFold [38], predicting 50 SA-generated and 50 stored sequences per family with the `alphafold2_ptm` model (single model, 3 recycles, no relaxation). For each family, we selected the SA-generated structure with the highest mean pLDDT and superimposed it on the experimental reference using PyMOL to produce the structural gallery in Fig. 5.

Antimicrobial peptide biophysical analysis. The beta defensin family (PF00711) provided an opportunity to assess whether generated sequences preserved biophysical properties associated with antimicrobial function. For each generated and stored sequence, we computed the net charge at pH 7 (summing +1 for Arg and Lys, -1 for Asp and Glu), the hydrophobic residue fraction (Ala, Val, Ile, Leu, Met, Phe, Trp, Pro), the mean Kyte–Doolittle hydrophobicity, and the cysteine count. We defined an antimicrobial plausibility filter requiring net charge $\geq +2$, hydrophobic ratio in $[0.3, 0.7]$, and ≥ 4 cysteines (reflecting the three disulfide bonds characteristic of the beta-defensin fold), and computed the pass rate for each method.

Critical temperature prediction. We pooled $n=33$ data points (the 8 Pfam families and 25 WW scaling replicates, 5 family sizes $\times$ 5 repeats) and fit ordinary least-squares regression models predicting β^* from MSA-derived features. We considered four candidate features: $\sqrt{d}$ (PCA dimensionality at 95% retained variance), mean per-column Shannon entropy $\bar{H}_{\text{col}}$, $\log K_{\text{eff}}$ (effective number of sequences at 80% identity), and spectral concentration $\lambda_1/\text{tr}(\mathbf{C})$ (leading eigenvalue of the sequence covariance, normalized by trace). We evaluated nested models by adding features in order of marginal R^2 improvement. Uncertainty on regression coefficients was quantified via nonparametric bootstrap: we resampled the 33 observations with replacement 10,000 times, refitting the model on each resample, and reported the standard deviation of the bootstrap distribution as the coefficient SE. Model comparison used root-mean-square error (RMSE) and R^2 , with the naive random-pattern prediction $\beta^* = \sqrt{d}$ as a reference baseline.

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Competing interests. The author declares no competing interest.

Data availability. All code, data, and analysis scripts are available at <https://github.com/varnerlab/SA-Protein-Modeling-Study>. Pfam seed alignments were obtained from InterPro [20]. Reference structures were obtained from the RCSB Protein Data Bank.

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Table 1: Properties of the eight Pfam families used in this study. K : number of seed sequences after cleaning. L : alignment length (columns retained). d : PCA dimension (95% variance). $\bar{H}_{\text{col}}$: mean per-column Shannon entropy (nats). K_{eff} : effective number of sequences at 80% identity threshold. β^* : empirical entropy inflection point. $\beta^*/\sqrt{d}$: ratio to the random-pattern prediction.

Family	Pfam ID	K	L	d	$\bar{H}_{\text{col}}$	K_{eff}	β^*	$\beta^*/\sqrt{d}$
RRM	PF00076	68	71	59	1.92	68	3.85	0.50
SH3	PF00018	55	48	46	1.68	55	3.85	0.57
WW	PF00397	420	31	186	1.74	419	5.45	0.40
Kunitz	PF00014	99	53	80	1.61	99	3.85	0.43
zf-C2H2	PF00096	151	23	106	1.91	151	4.58	0.44
PDZ	PF00595	44	83	37	1.88	43	3.23	0.53
Pkinase	PF00069	37	262	34	1.72	37	3.23	0.55
Defensin_beta	PF00711	45	36	37	1.65	45	3.23	0.53

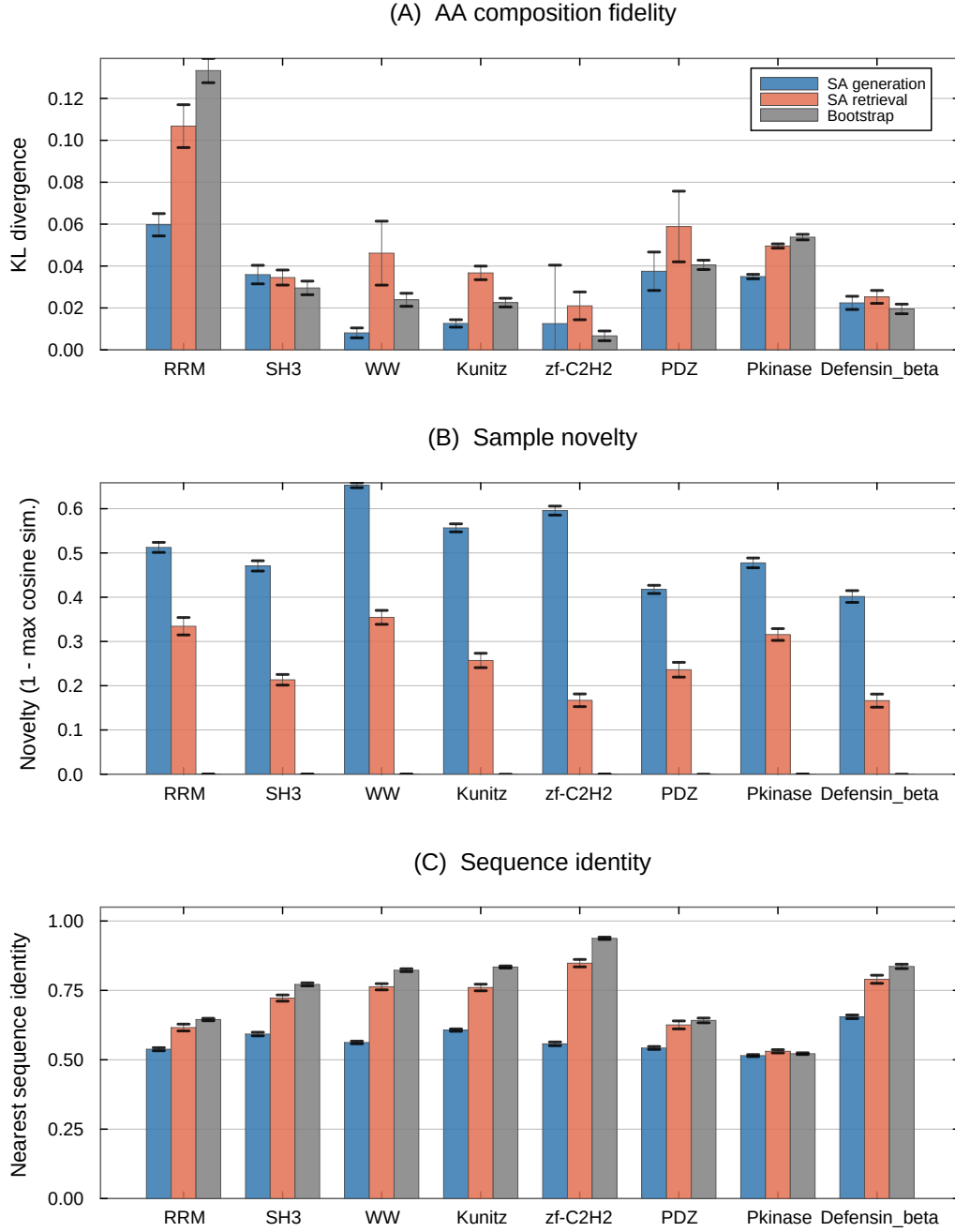


Figure 1: Cross-family comparison of SA generation, SA retrieval, and bootstrap across eight Pfam families. **(A)** KL divergence of amino acid composition (lower is better). SA generation achieves the lowest or near-lowest KL in every family. **(B)** Novelty in PCA space ($1 - \max_k \cos(\hat{\xi}, \mathbf{m}_k)$; higher is better). SA generation produces substantially novel sequences, while bootstrap samples, which are exact copies of stored patterns, have zero PCA-space novelty by construction. **(C)** Nearest sequence identity to a stored pattern (lower indicates greater departure). Error bars show ± 1 SE (30 chains $\times$ 5 samples).

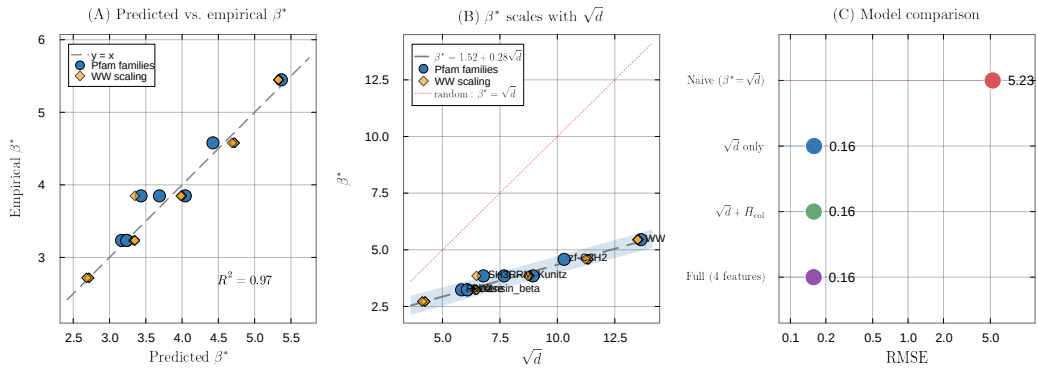


Figure 2: Predicting the critical temperature from MSA statistics. **(A)** Predicted vs. empirical β^* for the simple linear model $\beta^* \approx 1.57 + 0.28\sqrt{d}$ ($R^2=0.97$, $n=33$). Blue circles: eight Pfam families; gold diamonds: WW scaling replicates ($K \in \{20, \dots, 400\}$, five repeats each). **(B)** β^* as a function of $\sqrt{d}$, with the fitted regression (dashed) and the random-pattern prediction $\beta^* = \sqrt{d}$ (dotted red). The intercept offset and reduced slope reflect structured correlations in real protein families. **(C)** Model comparison (log-scale RMSE). The simple $\sqrt{d}$ model (RMSE=0.16) is $33\times$ more accurate than the naive random-pattern prediction $\beta^* = \sqrt{d}$ (RMSE=5.29); adding MSA features provides negligible improvement.

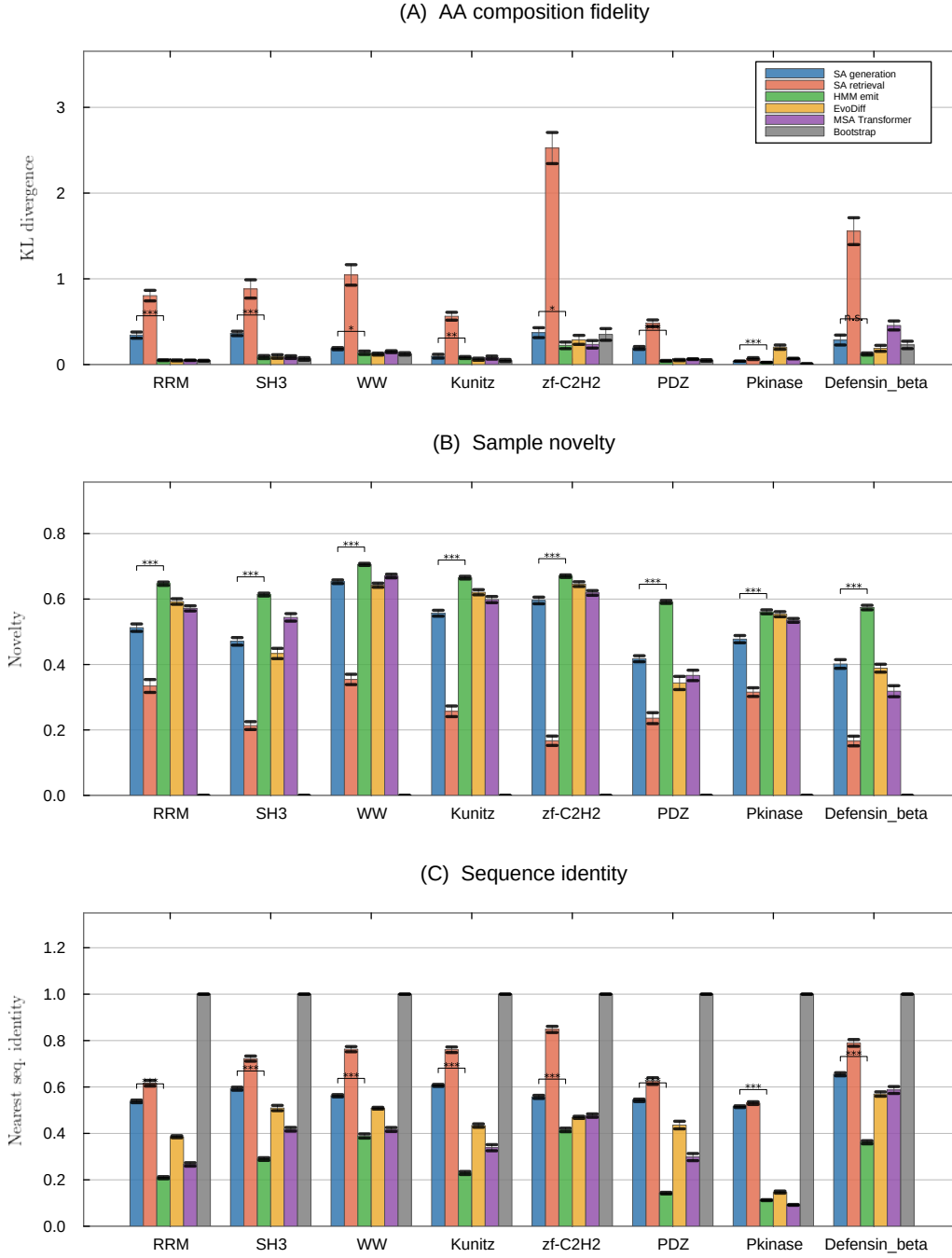


Figure 3: Head-to-head comparison of SA generation against learned baselines across all eight Pfam families. **(A)** KL divergence of amino acid composition. **(B)** Novelty in PCA space. **(C)** Nearest sequence identity. Significance brackets show Wilcoxon rank-sum tests comparing SA generation to HMM emit (the most direct comparison, as both use only the seed alignment): *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant. Error bars show ± 1 SE (30 chains $\times$ 5 samples).

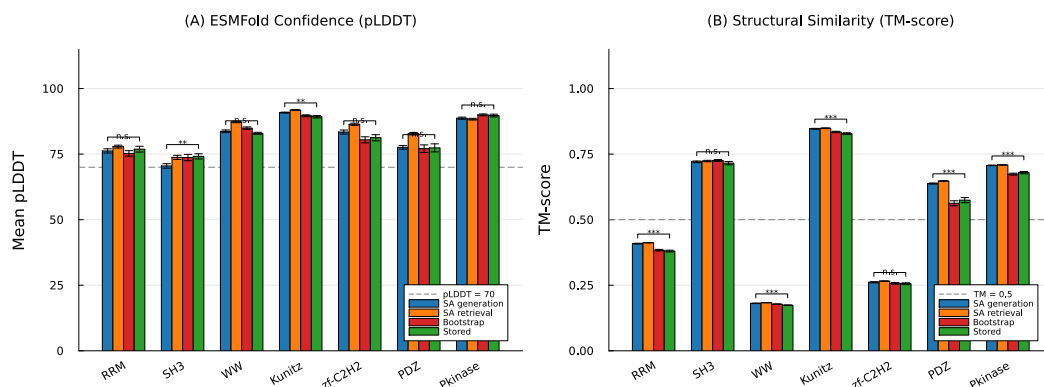


Figure 4: Structure validation of SA-generated sequences using ESMFold. **(A)** Mean pLDDT (folding confidence) across seven Pfam families (ESMFold predictions for Defensin_beta are pending) for SA generation, SA retrieval, bootstrap, and stored (natural) sequences. The dashed line indicates the pLDDT=70 confidence threshold. **(B)** TM-score to a representative experimentally determined structure for each family. The dashed line indicates TM=0.5 (same-fold threshold). Significance brackets show Wilcoxon rank-sum tests comparing SA generation to stored sequences: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant. Error bars show ± 1 SE ($n=50$ sequences per condition).

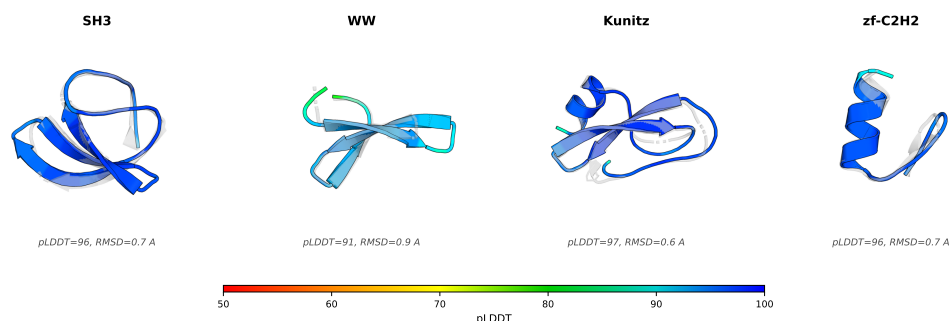


Figure 5: AlphaFold2 predicted structures for representative SA-generated sequences (colored by per-residue pLDDT confidence) superimposed on the experimentally determined reference structure for each family (gray, semi-transparent). For each family, the SA-generated sequence with the highest mean pLDDT was selected and aligned to the reference using PyMOL. The close structural agreement (sub-angstrom RMSD) and uniformly high pLDDT scores confirm that SA-generated sequences, despite substantial sequence-level novelty, encode three-dimensional folds that faithfully recapitulate the canonical architecture of their target family.

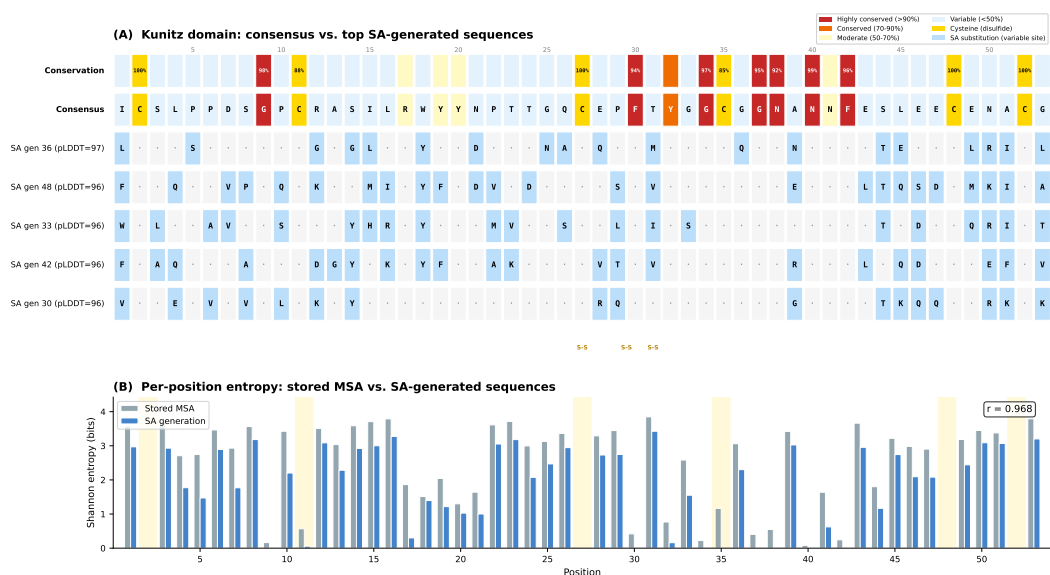


Figure 6: Sequence-level analysis of SA-generated Kunitz domain sequences. **(A)** Alignment of the family consensus with the five highest-confidence SA-generated sequences (ranked by AlphaFold2 pLDDT). Positions are colored by conservation in the stored MSA: highly conserved ($> 90\%$, red), conserved ($70\text{--}90\%$, orange), moderate ($50\text{--}70\%$, yellow), and variable ($< 50\%$, blue). Cysteines forming the three disulfide bonds are highlighted in gold. Dots indicate matches to the consensus; letters indicate substitutions. All five sequences preserve every highly conserved position (11/11) and all six cysteines (6/6), while introducing 17–25 substitutions concentrated at variable sites. **(B)** Per-position Shannon entropy for the stored MSA (gray) and SA-generated sequences (blue). The near-perfect correlation ($r=0.968$) indicates that SA faithfully recapitulates the position-specific conservation pattern of the family: structurally critical positions remain invariant, while evolutionarily permissive positions exhibit comparable diversity.

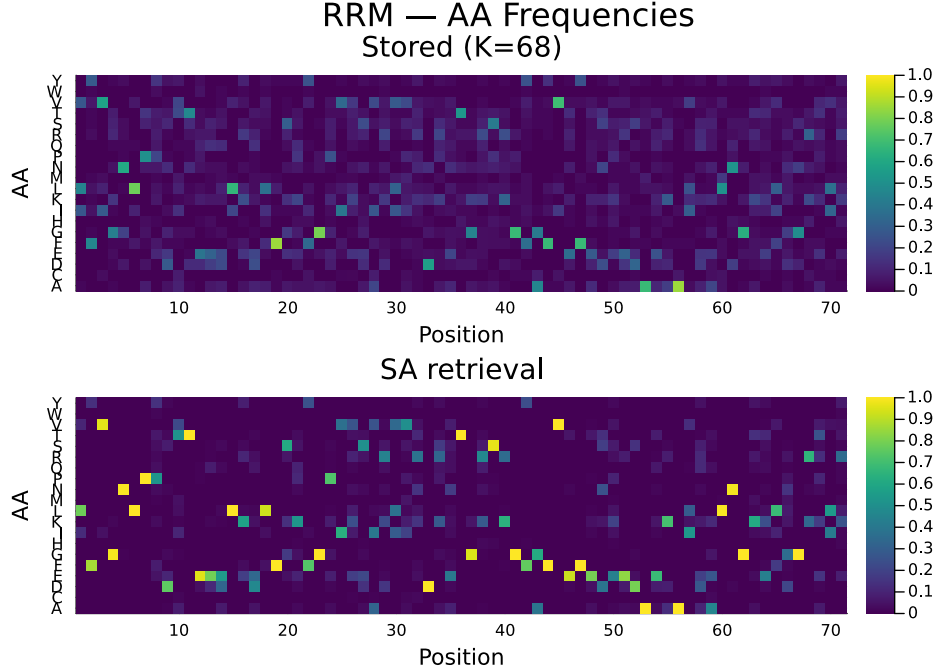


Figure S1: Per-position amino acid frequency profiles for the RRM family (PF00076). **Top:** stored seed sequences ($K=68$). **Bottom:** SA-generated sequences ($S=150$, retrieval regime). The generated ensemble reproduces the position-specific conservation pattern of the family, including strongly conserved positions and variable regions.

SI Appendix

S1 Amino Acid Composition Profiles

We compared the per-position amino acid frequency profiles of the stored seed sequences and SA-generated sequences for the RRM family (PF00076, Fig. S1). We found that the generated ensemble faithfully reproduces the conservation pattern of the family: strongly conserved positions (e.g., the aromatic residues characteristic of the ribonucleoprotein motifs RNP1 and RNP2) retain high frequency in the generated output, while variable positions display a similar breadth of amino acid usage. This agreement is consistent with the low KL divergences reported in Table S1 and shows that the Boltzmann sampling procedure captures not only global amino acid composition but also the position-specific constraints encoded in the seed alignment.

S2 Scaling Study

To characterize how generation quality degrades with decreasing family size, we subsampled the WW domain ($K=420$) at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition (Fig. S2). Stochastic attention in the generation regime maintained low KL divergence across the entire range (0.019 ± 0.008 at $K=20$, improving to 0.010 ± 0.002 at $K=400$), while the convex combination baseline exceeded $KL=0.8$ at every family size. Novelty increased with K (from 0.47 to 0.74), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56). The empirical critical temperature β^* increased from 2.7 at $K=20$ to 5.5 at $K=400$, tracking the growth in PCA dimensionality.

S3 Generation Metrics

Table S1 reports the full generation metrics for all eight Pfam families across five sampling conditions: SA generation ($\beta \approx 2\beta^*$), SA retrieval ($\beta \approx 20\beta^*$), bootstrap resampling, Gaussian perturbation,

and convex combination. For each condition, we report the KL divergence of amino acid composition, PCA-space cosine novelty, and nearest sequence identity to a stored pattern. Standard errors are computed over 30 independent Langevin chains of 5 samples each; KL standard errors are obtained via 1,000-fold bootstrap resampling. Bold entries indicate the best-performing method for each family and metric.

Table S1: Generation metrics across eight Pfam families (mean $\pm$ SE). SA gen: stochastic attention generation regime ($\beta \approx 2\beta^*$). SA ret: retrieval regime ($\beta \approx 20\beta^*$). BS: bootstrap. GP: Gaussian perturbation. CC: convex combination. KL: KL divergence of amino acid composition ($\downarrow$ lower is better; SE via 1,000-fold bootstrap). Novelty: $1 - \max_k \cos(\hat{\xi}, \mathbf{m}_k)$ in PCA space ($\uparrow$ higher is better). SeqID: nearest sequence identity to a stored pattern after decoding ($\downarrow$ lower is better, indicating greater novelty). Standard errors for novelty and SeqID are computed over 30 independent chains of 5 samples each.

Family	Method	KL _{AA} ($\downarrow$)	Novelty ($\uparrow$)	SeqID ($\downarrow$)
RRM	SA gen	0.060 $\pm$ 0.005	0.513 $\pm$ 0.011	0.538 $\pm$ 0.006
	SA ret	0.107 $\pm$ 0.010	0.334 $\pm$ 0.020	0.616 $\pm$ 0.012
	BS	0.133 $\pm$ 0.006	0.000 $\pm$ 0.000	0.645 $\pm$ 0.004
	GP	0.132 $\pm$ 0.005	0.008 $\pm$ 0.000	0.633 $\pm$ 0.006
	CC	2.091 $\pm$ 0.000	0.510 $\pm$ 0.009	0.562 $\pm$ 0.000
SH3	SA gen	0.036 $\pm$ 0.004	0.471 $\pm$ 0.012	0.593 $\pm$ 0.006
	SA ret	0.035 $\pm$ 0.004	0.213 $\pm$ 0.012	0.722 $\pm$ 0.011
	BS	0.030 $\pm$ 0.003	0.000 $\pm$ 0.000	0.772 $\pm$ 0.006
	GP	0.030 $\pm$ 0.003	0.006 $\pm$ 0.000	0.752 $\pm$ 0.006
	CC	0.798 $\pm$ 0.012	0.474 $\pm$ 0.008	0.601 $\pm$ 0.002
WW	SA gen	0.008 $\pm$ 0.002	0.653 $\pm$ 0.006	0.562 $\pm$ 0.005
	SA ret	0.046 $\pm$ 0.013	0.354 $\pm$ 0.016	0.763 $\pm$ 0.011
	BS	0.024 $\pm$ 0.003	0.000 $\pm$ 0.000	0.823 $\pm$ 0.005
	GP	0.029 $\pm$ 0.004	0.017 $\pm$ 0.000	0.809 $\pm$ 0.006
	CC	1.375 $\pm$ 0.092	0.637 $\pm$ 0.004	0.766 $\pm$ 0.001
Kunitz	SA gen	0.013 $\pm$ 0.002	0.557 $\pm$ 0.009	0.607 $\pm$ 0.004
	SA ret	0.037 $\pm$ 0.003	0.257 $\pm$ 0.016	0.760 $\pm$ 0.012
	BS	0.023 $\pm$ 0.002	0.000 $\pm$ 0.000	0.834 $\pm$ 0.004
	GP	0.024 $\pm$ 0.002	0.010 $\pm$ 0.000	0.844 $\pm$ 0.004
	CC	0.721 $\pm$ 0.013	0.546 $\pm$ 0.007	0.590 $\pm$ 0.001
zf-C2H2	SA gen	0.013 $\pm$ 0.028	0.596 $\pm$ 0.010	0.557 $\pm$ 0.007
	SA ret	0.021 $\pm$ 0.010	0.167 $\pm$ 0.014	0.848 $\pm$ 0.013
	BS	0.007 $\pm$ 0.002	0.000 $\pm$ 0.000	0.938 $\pm$ 0.004
	GP	0.007 $\pm$ 0.002	0.011 $\pm$ 0.000	0.940 $\pm$ 0.004
	CC	2.346 $\pm$ 0.060	0.580 $\pm$ 0.006	0.600 $\pm$ 0.001
PDZ	SA gen	0.038 $\pm$ 0.010	0.418 $\pm$ 0.009	0.543 $\pm$ 0.005
	SA ret	0.059 $\pm$ 0.016	0.236 $\pm$ 0.017	0.626 $\pm$ 0.014
	BS	0.041 $\pm$ 0.002	0.000 $\pm$ 0.000	0.642 $\pm$ 0.009
	GP	0.045 $\pm$ 0.003	0.006 $\pm$ 0.000	0.648 $\pm$ 0.007
	CC	0.339 $\pm$ 0.001	0.449 $\pm$ 0.009	0.549 $\pm$ 0.001
Pkinase	SA gen	0.035 $\pm$ 0.001	0.478 $\pm$ 0.011	0.515 $\pm$ 0.004
	SA ret	0.050 $\pm$ 0.001	0.316 $\pm$ 0.013	0.531 $\pm$ 0.006
	BS	0.054 $\pm$ 0.001	0.000 $\pm$ 0.000	0.522 $\pm$ 0.003
	GP	0.051 $\pm$ 0.001	0.005 $\pm$ 0.000	0.533 $\pm$ 0.003
	CC	0.062 $\pm$ 0.001	0.446 $\pm$ 0.009	0.470 $\pm$ 0.000
Defensin_beta	SA gen	0.022 $\pm$ 0.004	0.402 $\pm$ 0.013	0.655 $\pm$ 0.007
	SA ret	0.025 $\pm$ 0.008	0.166 $\pm$ 0.015	0.790 $\pm$ 0.015
	BS	0.020 $\pm$ 0.003	0.000 $\pm$ 0.000	0.836 $\pm$ 0.006
	GP	0.023 $\pm$ 0.003	0.006 $\pm$ 0.000	0.835 $\pm$ 0.006
	CC	0.708 $\pm$ 0.001	0.442 $\pm$ 0.009	0.716 $\pm$ 0.001

S4 Independent Fitness Validation via ESM2

As an independent assessment of whether SA-generated sequences resemble natural proteins, we scored all generated and stored sequences using ESM2-650M [31], a protein language model pre-trained on millions of UniRef50 sequences that is entirely independent of the stochastic attention framework. For each sequence, we computed the pseudo-perplexity via masked marginal scoring: each position was masked in turn, the model predicted a distribution over amino acids, and the log-probability of the true residue was recorded. The pseudo-perplexity (lower is more protein-like) is $\exp(-\frac{1}{L} \sum_{\ell=1}^L \log P(a_{\ell} | \mathbf{a}_{\setminus \ell}))$, where a_{ℓ} is the amino acid at position ℓ and $\mathbf{a}_{\setminus \ell}$ denotes all other positions. Wilcoxon rank-sum tests with Cohen’s d effect sizes compared SA-generated sequences ($n=150$) to stored natural sequences ($n=37-50$) for each family (Table S2).

SA-generated sequences were statistically indistinguishable from natural family members in five of eight families (Wilcoxon $p > 0.05$; RRM, zf-C2H2, Pkinase, Defensin_beta: $p > 0.29$). In the Kunitz domain, SA generation achieved significantly *lower* (better) pseudo-perplexity than stored sequences (4.47 vs. 4.98, $p < 0.001$, $d=-0.74$). Three families showed modestly higher pseudo-perplexity for SA generation (SH3: $d=0.60$; WW: $d=1.03$; PDZ: $d=0.60$), though all SA-generated sequences remained well within the range of natural protein sequences. The overall mean pseudo-perplexity across all families was 5.71 for SA-generated sequences versus 5.58 for stored sequences, confirming that an independent protein language model trained on billions of sequences considers SA-generated sequences to be comparably protein-like to natural family members.

Table S2: ESM2-650M pseudo-perplexity validation. Lower pseudo-perplexity indicates a more protein-like sequence. p : Wilcoxon rank-sum test (two-sided). d : Cohen’s effect size (negative values indicate SA is better). Significance: *** $p < 0.001$, ** $p < 0.01$, n.s. = not significant ($p > 0.05$).

Family	SA gen ppl	Stored ppl	p	d	Sig.
RRM	4.87 $\pm$ 0.68	5.20 $\pm$ 1.92	0.30	-0.29	n.s.
SH3	5.24 $\pm$ 0.82	4.62 $\pm$ 1.48	< 0.001	0.60	***
WW	6.78 $\pm$ 1.77	5.04 $\pm$ 1.36	< 0.001	1.03	***
Kunitz	4.47 $\pm$ 0.57	4.98 $\pm$ 0.94	< 0.001	-0.74	***
zf-C2H2	7.26 $\pm$ 1.66	8.10 $\pm$ 3.47	0.41	-0.37	n.s.
PDZ	6.30 $\pm$ 1.21	5.31 $\pm$ 2.61	0.002	0.60	**
Pkinase	4.44 $\pm$ 0.32	4.32 $\pm$ 0.80	0.39	0.27	n.s.
Defensin_beta	6.35 $\pm$ 1.27	6.82 $\pm$ 2.09	0.32	-0.31	n.s.

S5 Deep Mutational Scanning Validation

To assess whether SA-generated substitutions correspond to experimentally tolerated mutations, we cross-referenced the amino acid changes introduced by SA generation against published deep mutational scanning (DMS) datasets for two families: the PDZ domain (DLG4/PSD-95 PDZ3; McLaughlin et al. [33]) and the SH3 domain (GRB2 SH3; Faure et al. [34]). For each family, we identified the DMS reference protein in the Pfam seed alignment (DLG4_RAT/313–391 for PDZ, GRB2_CHICK/162–207 for SH3), mapped alignment positions to DMS coordinates, and classified each SA-generated substitution relative to the family consensus as experimentally tolerated (DMS_score_bin=1) or deleterious (DMS_score_bin=0) based on the published fitness data (Table S3).

SA-generated substitutions were strongly enriched for experimentally tolerated mutations in both families. In the PDZ domain, 91.7% of the 4,313 matched substitutions were classified as tolerated, compared to an overall tolerance rate of 77.5% in the DMS dataset (1.18 $\times$ enrichment, binomial $p < 10^{-135}$). The continuous DMS fitness scores of SA-generated substitutions were also significantly higher than those of random single mutations (Mann-Whitney $p < 10^{-21}$, Cohen’s $d=0.42$). In the SH3 domain, the enrichment was even stronger: 90.4% of 2,396 matched substitutions were tolerated versus 67.5% overall (1.34 $\times$ enrichment, binomial $p < 10^{-156}$), with a larger effect size on continuous fitness scores ($d=0.70$, $p < 10^{-55}$). These results demonstrate that the Boltzmann sampling procedure preferentially generates substitutions at positions and to residues that have been experimentally validated as functional, without any explicit fitness objective or training signal.

Table S3: Deep mutational scanning validation. SA-generated substitutions relative to the family consensus were matched to published single-mutant DMS fitness data. Tolerance rate: fraction of matched substitutions classified as functional (DMS_score_bin=1). DMS overall: baseline tolerance rate across all single mutations in the dataset. Enrichment: ratio of SA tolerance rate to DMS overall rate. p : one-sided binomial test (H_0 : SA tolerance rate $\leq$ DMS overall rate).

Family	DMS source	Matched	SA tol. rate	DMS overall	Enrichment	p
PDZ	McLaughlin et al. 2012	4,313	91.7%	77.5%	1.18 $\times$	$< 10^{-135}$
SH3	Faure et al. 2022	2,396	90.4%	67.5%	1.34 $\times$	$< 10^{-156}$

S6 Potts Model (plmDCA) Baseline

To assess whether pairwise coevolutionary models provide a stronger baseline than profile HMMs, we fitted a Potts model via pseudo-likelihood maximization (plmDCA) [8] to the SH3 and Kunitz seed alignments and generated 150 sequences per family by Gibbs sampling (500 sweeps per chain, initialized from random stored sequences). The Potts model captures pairwise couplings $J_{ij}(a, b)$ between all position pairs in addition to single-site fields $h_i(a)$, giving it substantially more parameters than a profile HMM. For SH3 ($K=55$, $L=48$), the model has approximately 498,000 parameters for 55 training sequences (data-to-parameter ratio $\approx 10^{-4}$); for Kunitz ($K=99$, $L=53$), approximately 609,000 parameters for 99 sequences. Despite this extreme over-parameterization, L2 regularization ($\lambda_J=\lambda_h=0.01$) yielded well-behaved generation (Table S4).

The Potts model achieved comparable or slightly better amino acid composition fidelity than SA (KL divergence 0.025 vs. 0.036 for SH3; 0.010 vs. 0.013 for Kunitz) while producing sequences with modestly lower novelty (0.37 vs. 0.47 for SH3; 0.54 vs. 0.56 for Kunitz) and lower nearest sequence identity (0.54 vs. 0.59 for SH3; 0.55 vs. 0.61 for Kunitz). Both methods occupy the same intermediate regime between profile HMMs (which drift far from the family at 23–29% identity) and bootstrap resampling (which has zero novelty). The critical distinction is computational: the Potts model required fitting $\sim 500,000$ parameters via L-BFGS-B optimization at each of L positions, followed by 500 Gibbs sweeps per sample (total wall time: ~ 100 s per family), while SA required no fitting and generated sequences in seconds. The similar output quality supports the conclusion that the information captured by pairwise couplings is also present in the Hopfield energy landscape, which implicitly encodes all-order correlations through the softmax attention operation.

Table S4: Potts model (plmDCA) baseline comparison for two Pfam families. Metrics are computed identically to the main baseline analysis (30 chains $\times$ 5 samples). KL: amino acid composition KL divergence ($\downarrow$). Novelty: PCA-space cosine distance ($\uparrow$). SeqID: nearest sequence identity to a stored pattern. Data/param: ratio of training sequences to model parameters.

Family	Method	KL _{AA} ($\downarrow$)	Novelty ($\uparrow$)	SeqID	Data/param
SH3	SA generation	0.036 $\pm$ 0.026	0.471 $\pm$ 0.012	0.593 $\pm$ 0.007	—
	Potts (plmDCA)	0.025 $\pm$ 0.019	0.374 $\pm$ 0.011	0.539 $\pm$ 0.009	1.1 $\times 10^{-4}$
	HMM emit	0.009 $\pm$ 0.017	0.614 $\pm$ 0.004	0.290 $\pm$ 0.006	—
Kunitz	SA generation	0.013 $\pm$ 0.022	0.557 $\pm$ 0.009	0.607 $\pm$ 0.004	—
	Potts (plmDCA)	0.010 $\pm$ 0.017	0.540 $\pm$ 0.010	0.548 $\pm$ 0.010	1.6 $\times 10^{-4}$
	HMM emit	0.030 $\pm$ 0.011	0.666 $\pm$ 0.004	0.229 $\pm$ 0.007	—

S7 Pairwise Covariation (Mutual Information) Analysis

To assess whether SA-generated sequences preserve pairwise residue covariation, we computed mutual information (MI) matrices for the stored seed alignment and for sequences generated by each method. For each family, we constructed the $L \times L$ MI matrix, where position pair (i, j) has MI defined as $\text{MI}(i, j) = \sum_{a,b} p(a, b) \log \frac{p(a, b)}{p_i(a) p_j(b)}$, with an additive pseudocount of 1.0 to regularize sparse counts. We then extracted the upper-triangle elements ($L(L-1)/2$ pairs) and computed Pearson and Spearman correlations between the stored MI vector and the MI vector of each generated

ensemble. We also computed the fraction of the top 50 highest-MI pairs in the stored alignment that appeared among the top 50 pairs in the generated ensemble.

Table S5 reports the MI correlation for SA generation and all baselines across the eight families. The central result is that SA-generated sequences preserve the pairwise covariation structure of the stored alignment: Pearson $r=0.53\text{--}0.92$ in seven of eight families (mean $r=0.72$ excluding Pkinase), with SA consistently outperforming profile HMM sequences (mean $r=0.42$), which model each position independently. The Pkinase family ($K=37$, $L=262$) was an outlier for all methods, consistent with the extreme data-to-parameter ratio limiting MI estimation quality in both the stored and generated ensembles. Methods that explicitly model pairwise couplings (Potts/plmDCA: $r=0.75\text{--}0.89$) or leverage pretraining on millions of MSAs (EvoDiff: $r=0.59\text{--}0.97$) achieved higher MI correlations, as expected given their direct access to pairwise or higher-order statistics. The ordering $\text{HMM} < \text{SA} < \text{Potts} \leq \text{EvoDiff}$ is consistent with the information each method encodes: position-independent frequencies, the implicit all-order couplings of the Hopfield energy, explicitly fitted pairwise couplings, and deep coevolutionary priors from large-scale pretraining, respectively. Figure S3 shows MI scatter plots for three representative families (Kunitz, SH3, Defensin_beta), with structural contact pairs highlighted in red. Contact pairs follow the same correlation trend as non-contact pairs, confirming that SA preserves covariation at structurally relevant positions.

Table S5: Pairwise mutual information (MI) preservation: Pearson correlation r between upper-triangle MI vectors of stored sequences and sequences generated by each method. Higher values indicate better preservation of pairwise residue covariation. Potts model results are shown only for the two families where plmDCA sequences were generated.

Family	SA gen	SA ret	Potts	HMM	EvoDiff	MSAT
RRM	0.53	0.34	—	0.49	0.86	0.60
SH3	0.66	0.64	0.75	0.46	0.88	0.73
WW	0.71	0.49	—	0.29	0.83	0.58
Kunitz	0.80	0.69	0.89	0.32	0.87	0.50
zf-C2H2	0.92	0.87	—	0.72	0.97	0.92
PDZ	0.59	0.44	—	0.07	0.81	0.01
Pkinase	0.05	−0.09	—	0.23	0.59	0.05
Defensin_beta	0.85	0.75	—	0.83	0.95	0.88

S8 Sequence Encoding, PCA Projection, and Decoding

Stochastic attention operates on continuous vectors on the unit sphere $\mathbb{S}^{d-1}$, while protein sequences are discrete objects over a 20-letter amino acid alphabet. Bridging the two requires three transformations: one-hot encoding, PCA dimensionality reduction, and unit-norm projection. Each step serves a specific purpose in the pipeline, and the reverse path (inverse PCA followed by argmax decoding) recovers discrete sequences from continuous samples. We describe each step in detail, state the mathematical formulation, and explain why PCA is the appropriate choice for this framework.

One-hot encoding. Given a cleaned alignment of K sequences, each of length L amino acids over the standard 20-letter alphabet $\mathcal{A} = \{\text{A}, \text{R}, \text{N}, \dots, \text{V}\}$, we encode each sequence as a binary vector in $\mathbb{R}^{20L}$ by concatenating per-position indicator vectors. Specifically, for sequence k with amino acid $a_\ell^{(k)}$ at position ℓ , the one-hot representation is

$$\mathbf{x}_k = [\mathbf{e}_{a_1^{(k)}}, \mathbf{e}_{a_2^{(k)}}, \dots, \mathbf{e}_{a_L^{(k)}}]^\top \in \{0, 1\}^{20L}, \quad (7)$$

where $\mathbf{e}_a \in \{0, 1\}^{20}$ is the standard basis vector for amino acid a . The full-dimensional representation is $d_{\text{full}} = 20L$, which ranges from 460 (zf-C2H2, $L=23$) to 5,240 (Pkinase, $L=262$) across the eight families studied. This encoding is lossless and treats each position-amino acid pair as an independent coordinate, making no assumptions about residue similarity or physicochemical properties.

Why dimensionality reduction is necessary. The full one-hot space is vastly over-dimensional relative to the number of stored patterns. For the families studied here, the ratio d_{full}/K ranges from 1.5 (WW, $K=420$, $d_{\text{full}}=620$) to 142 (Pkinase, $K=37$, $d_{\text{full}}=5,240$). This creates two problems for

stochastic attention. First, in the modern Hopfield energy, the similarity scores $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ have variance $\text{Var}(e_k) = 1/d$ for a random query on $\mathbb{S}^{d-1}$ (see Appendix S9). When d is large, the scores concentrate tightly around zero, and extremely high β is required to differentiate among stored patterns, which makes the energy landscape nearly flat over most of the sphere and renders Langevin sampling inefficient. Second, the one-hot encoding introduces massive redundancy: at each position, only 1 of 20 coordinates is nonzero, and the per-column sum constraint means that each position contributes at most 19 degrees of freedom (and often fewer when positions are strongly conserved). PCA removes this redundancy by projecting onto the directions of actual variation in the data.

PCA projection. We compute PCA from the one-hot encoded alignment. Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_K] \in \mathbb{R}^{20L \times K}$ be the matrix of one-hot encoded sequences. We first center the data:

$$\bar{\mathbf{x}} = \frac{1}{K} \sum_{k=1}^K \mathbf{x}_k, \quad \tilde{\mathbf{X}} = \mathbf{X} - \bar{\mathbf{x}} \mathbf{1}_K^\top, \quad (8)$$

where $\bar{\mathbf{x}} \in \mathbb{R}^{20L}$ is the column mean (i.e., the family’s position-specific amino acid frequency profile) and $\mathbf{1}_K$ is the K -vector of ones. We then compute the economy singular value decomposition (SVD)

$$\tilde{\mathbf{X}} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\top, \quad (9)$$

where $\mathbf{U} \in \mathbb{R}^{20L \times r}$ contains the left singular vectors (principal directions), $\boldsymbol{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_r)$ with $\sigma_1 \geq \dots \geq \sigma_r > 0$ are the singular values, $\mathbf{V} \in \mathbb{R}^{K \times r}$ contains the right singular vectors, and $r = \text{rank}(\tilde{\mathbf{X}}) \leq \min(20L, K) - 1$. The variance captured by the j -th principal component is σ_j^2/K , and the fraction of total variance explained by the first p components is

$$\rho(p) = \frac{\sum_{j=1}^p \sigma_j^2}{\sum_{j=1}^r \sigma_j^2}. \quad (10)$$

We select p as the smallest integer such that $\rho(p) \geq 0.95$, retaining 95% of the variance. The projection matrix is $\mathbf{W}_p = \mathbf{U}_{:,1:p} \in \mathbb{R}^{20L \times p}$ (the first p columns of $\mathbf{U}$), and the projected representation of each sequence is

$$\mathbf{z}_k = \mathbf{W}_p^\top (\mathbf{x}_k - \bar{\mathbf{x}}) \in \mathbb{R}^p. \quad (11)$$

Among all rank- p linear projections, PCA minimizes the mean squared reconstruction error $\frac{1}{K} \sum_k \|\mathbf{x}_k - (\bar{\mathbf{x}} + \mathbf{W}_p \mathbf{z}_k)\|^2$, making it the optimal linear dimensionality reduction for preserving the structure of the data. Across the eight families, the retained PCA dimension $d \equiv p$ ranged from 34 (Pkinase) to 186 (WW), yielding compression ratios d_{full}/d of $3.3\times$ (WW) to $154\times$ (Pkinase). The large compression ratios for small families reflect the fact that few sequences span few independent directions in the full one-hot space.

Unit-norm projection. After PCA, we normalize each projected vector to unit L_2 norm:

$$\mathbf{m}_k = \frac{\mathbf{z}_k}{\|\mathbf{z}_k\|_2} \in \mathbb{S}^{d-1}. \quad (12)$$

This step places all stored patterns on the unit sphere, which is required by the modern Hopfield energy. The normalization ensures that the similarity scores $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ lie in $[-1, 1]$ and that the concentration-of-measure result $\text{Var}(e_k) = 1/d$ holds exactly for random queries (Appendix S9). It also prevents patterns with larger PCA norms (which correspond to sequences that deviate more from the family mean) from dominating the softmax attention weights purely by virtue of their norm rather than their direction. The columns $\mathbf{m}_1, \dots, \mathbf{m}_K$ form the memory matrix $\hat{\mathbf{X}} \in \mathbb{R}^{d \times K}$ used throughout the paper.

Decoding: from continuous samples to amino acid sequences. The Langevin sampler produces continuous vectors $\boldsymbol{\xi} \in \mathbb{R}^d$ in PCA space. To decode a sample back to an amino acid sequence, we apply two steps. First, the inverse PCA reconstruction:

$$\hat{\mathbf{x}} = \bar{\mathbf{x}} + \mathbf{W}_p \boldsymbol{\xi} \in \mathbb{R}^{20L}, \quad (13)$$

which maps the d -dimensional sample back to the full $20L$ -dimensional one-hot space. The result $\hat{\mathbf{x}}$ is a continuous vector that approximates a one-hot encoding but is not exactly binary. Second, we

apply per-position argmax decoding: for each position $\ell = 1, \dots, L$, we extract the 20-dimensional sub-vector $\hat{\mathbf{x}}_{[20(\ell-1)+1:20\ell]}$ and assign the amino acid with the largest coordinate:

$$a_\ell = \arg \max_{a \in \mathcal{A}} \hat{x}_{20(\ell-1)+a}. \quad (14)$$

This decoding is deterministic and produces a valid amino acid at every position. In practice, we found that 100% of decoded positions yielded standard amino acids across all families and conditions (Table S1, ValidAA column), indicating that the PCA representation and Langevin dynamics remain in a region of continuous space where the argmax decoding is unambiguous.

Why PCA rather than other representations? Several properties make PCA particularly well suited for use with the modern Hopfield energy. First, PCA is *linear*: both the forward projection $\mathbf{z} = \mathbf{W}_p^\top (\mathbf{x} - \bar{\mathbf{x}})$ and the inverse $\hat{\mathbf{x}} = \bar{\mathbf{x}} + \mathbf{W}_p \mathbf{z}$ are matrix multiplications, which means the score function $\nabla \log p_\beta(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi})$ remains an exact closed-form expression over the continuous PCA space with no approximation error from the representation itself. Nonlinear embeddings (e.g., autoencoders, UMAP) would break this property because the Hopfield energy would be defined over the latent space while the score function in the ambient space would require backpropagation through the decoder, reintroducing the training dependence that stochastic attention is designed to avoid. Second, PCA is *variance-optimal*: it captures the maximum amount of family-level variation in the fewest dimensions, which concentrates the discriminative structure of the family into a compact space where the Boltzmann distribution is well-conditioned. Third, PCA is *deterministic and parameter-free*: given the alignment, the projection is uniquely defined by the SVD and the variance threshold, with no hyperparameters to tune or training procedures that could overfit. This aligns with the training-free philosophy of the method. Finally, the resulting representation on $\mathbb{S}^{d-1}$ satisfies the concentration-of-measure properties (variance $1/d$, sub-Gaussian tails) that underpin the analytical framework for the critical temperature (Appendix S9).

S9 Analysis of the Critical Temperature

We developed an analytical framework to understand the empirical relationship $\beta^* \approx 1.57 + 0.28\sqrt{d}$ reported in Materials and Methods, starting from exact results on similarity variance, building a Gaussian mean-field prediction, and then diagnosing why the mean-field coefficient overshoots and how stored-pattern self-similarity resolves the discrepancy.

After one-hot encoding and PCA projection to d dimensions, we normalized the stored patterns $\mathbf{m}_k$ to lie on $\mathbb{S}^{d-1}$. For a random query $\boldsymbol{\xi}$ drawn uniformly from $\mathbb{S}^{d-1}$, each similarity $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ has mean zero and variance $1/d$, regardless of the structure of $\mathbf{m}_k$. This follows from the rotational invariance of the uniform distribution on $\mathbb{S}^{d-1}$: for any fixed unit vector $\mathbf{m}$,

$$\mathbb{E}[\mathbf{m}^\top \boldsymbol{\xi}] = 0, \quad \text{Var}(\mathbf{m}^\top \boldsymbol{\xi}) = \frac{1}{d}. \quad (15)$$

We verified this numerically for all eight protein families using 1,000 random probes per family. We found that the ratio $\sigma^2 \cdot d$ is 1.000 ± 0.003 across all families and WW subsamples, confirming that PCA-projected protein patterns behave identically to random unit vectors with respect to similarity variance (Fig. S4B,F). We also observed that the excess kurtosis is mildly negative (-0.06 to -0.33), consistent with the sub-Gaussian behavior expected on $\mathbb{S}^{d-1}$. These observations motivated a Gaussian mean-field model: if the K similarity scores were i.i.d. draws from $\mathcal{N}(0, \sigma^2)$ with $\sigma = 1/\sqrt{d}$, we can work entirely in dimensionless units by writing each score as $e_k = \sigma z_k$ with $z_k \sim \mathcal{N}(0, 1)$.

In these units the softmax entropy $H(\tau) = -\sum_k p_k \log p_k$ with $p_k \propto \exp(\tau z_k)$ undergoes a transition from $H = \log K$ (uniform, $\tau = 0$) to $H \rightarrow 0$ ($\tau \rightarrow \infty$). We defined the *dimensionless inflection* τ^* as the τ that maximizes $|d^2 H/d(\log \tau)^2|$ and computed it numerically by averaging over 500 Gaussian realizations for each $K \in \{10, 20, \dots, 1000\}$. We found that τ^* is nearly constant at ≈ 1.6 for $K \geq 30$, with only weak logarithmic growth (Fig. S4A). Translating back to physical units, the Gaussian prediction for the entropy inflection is $\beta^* = \tau^*/\sigma = \tau^*\sqrt{d} \approx 1.6\sqrt{d}$, which has the correct $\sqrt{d}$ scaling and achieves a Pearson correlation of $r = 0.98$ with the empirical values across all families and subsamples (Fig. S4C).

Despite the strong correlation, we found that the Gaussian prediction systematically overpredicts β^* by a factor of ≈ 4 ($R^2 = -180$), making it useless as an absolute predictor. We fitted the relationship

$$\beta^* = 1.565 + 0.281/\sigma = 1.565 + 0.281\sqrt{d} \quad (16)$$

and obtained $R^2 = 0.962$, with an intercept that matches τ^* to within numerical precision but a slope of 0.28, roughly $6\times$ smaller than $\tau^* \approx 1.6$. To understand this gap, we examined how β^* is actually computed. We evaluate the empirical inflection by probing at *stored patterns* $\mathbf{m}_k$, not at random queries on $\mathbb{S}^{d-1}$. When $\boldsymbol{\xi} = \mathbf{m}_k$, the similarity vector acquires a distinctive structure:

$$e_j = \mathbf{m}_j^\top \mathbf{m}_k = \begin{cases} 1 & j = k \\ \text{cross-similarity} & j \neq k, \end{cases} \quad (17)$$

since $\|\mathbf{m}_k\| = 1$. The self-similarity score of 1 is an outlier relative to the cross-similarities $\mathbf{m}_j^\top \mathbf{m}_k$ for $j \neq k$, which have magnitudes $\mathcal{O}(1/\sqrt{d})$. We measured the resulting gap $\Delta = 1 - \max_{j \neq k} \mathbf{m}_j^\top \mathbf{m}_k$ and found that it ranges from 0.55 to 0.82 across families (Fig. S5E), which shows that the softmax concentrates on the self-pattern at much lower β than would be needed for a random query where no score is pinned at the maximum of $\mathbb{S}^{d-1}$.

We then asked whether the Gaussian τ^*/σ framework could be salvaged by substituting the cross-similarity standard deviation σ_{cross} for $\sigma = 1/\sqrt{d}$, but we found that this fails decisively ($R^2 = -9.0$; Fig. S5D). We observed that the cross-similarity variance does not follow the $1/d$ law — it ranges from $0.19\times$ to $2.4\times$ the random-probe variance depending on the family’s correlation structure (Fig. S5B,C) — and the excess kurtosis is large (6–25), reflecting a heavy-tailed distribution driven by a few highly similar pattern pairs amid a bulk of near-orthogonal ones. The Gaussian mean-field decomposition $\beta^* = \tau^*/\sigma$ thus breaks down completely for stored-pattern probes: the self-similarity outlier and the non-Gaussian cross-similarity tails violate both assumptions simultaneously. Nonetheless, we found that the fitted model $\beta^* = 1.57 + 0.28\sqrt{d}$ remains robust. The $\sqrt{d}$ scaling is set by concentration of measure, which holds for any probe, and the self-similarity effect enters primarily through a reduced effective coefficient rather than a change in functional form. The intercept $\approx \tau^*$ recovers the universal softmax baseline, while the coefficient 0.28 encapsulates the combined effect of self-similarity anchoring and the protein family correlation structure — a quantity that is empirically stable across the eight families and five scaling conditions we studied, but for which we do not currently have a closed-form expression.

To quantify uncertainty in the fitted regression $\beta^* = 1.57 + 0.28\sqrt{d}$, we performed a nonparametric bootstrap analysis. We resampled the 33 observations (8 Pfam families and 25 WW scaling replicates) with replacement 10,000 times and refit the ordinary least-squares (OLS) model on each resample. We obtained a bootstrap SE of 0.07 for the intercept and 0.007 for the slope, yielding 95% confidence intervals of $[1.39, 1.68]$ and $[0.266, 0.294]$, respectively. We found that the bootstrap distributions of both coefficients are unimodal and approximately Gaussian, with no evidence of instability or sensitivity to individual data points. The bootstrap R^2 distribution has a median of 0.969 with a 95% interval of $[0.940, 0.987]$, confirming that the strong predictive relationship is not driven by outliers. We also verified that removing any single family from the dataset changes the fitted slope by less than 0.02, which shows that no individual family exerts undue leverage on the regression.

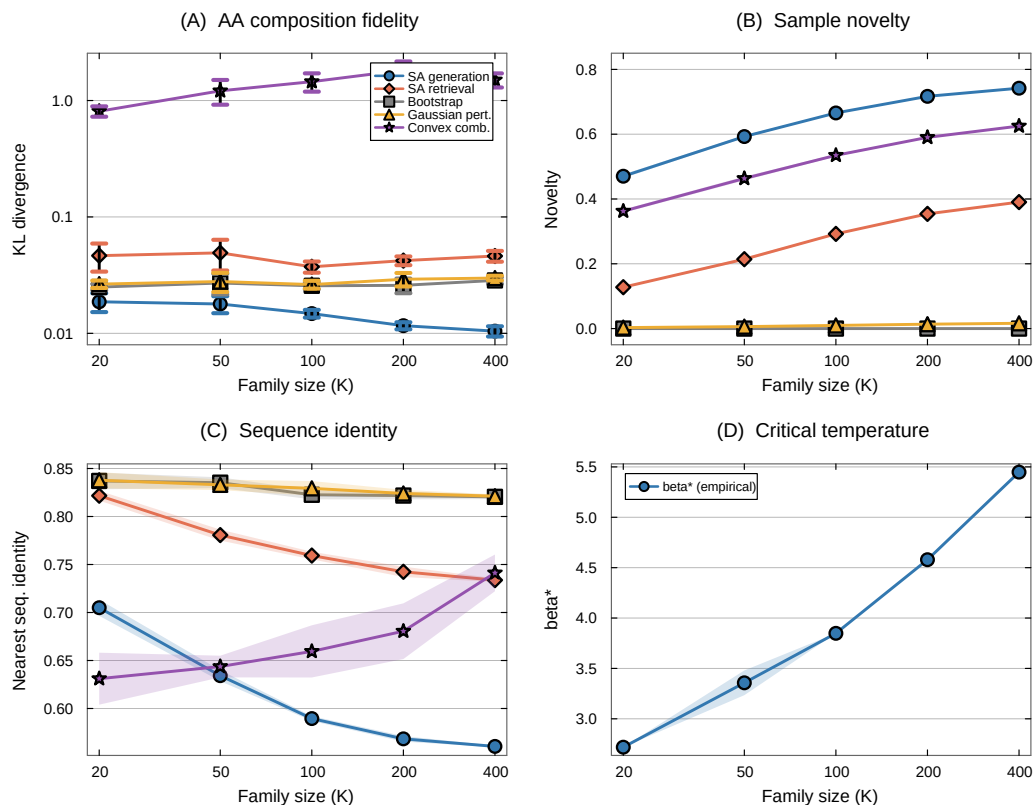


Figure S2: Scaling study: WW domain subsampled at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates. **(A)** KL divergence (log-log scale). SA generation maintains low KL across the entire range; convex combination degrades catastrophically. **(B)** Novelty increases with K for SA methods; baselines remain near zero. **(C)** Nearest sequence identity decreases with K , indicating the sampler explores further from stored patterns as more patterns shape the energy landscape. **(D)** The empirical critical temperature β^* increases with family size. Shaded regions and error bars show ± 1 SE.

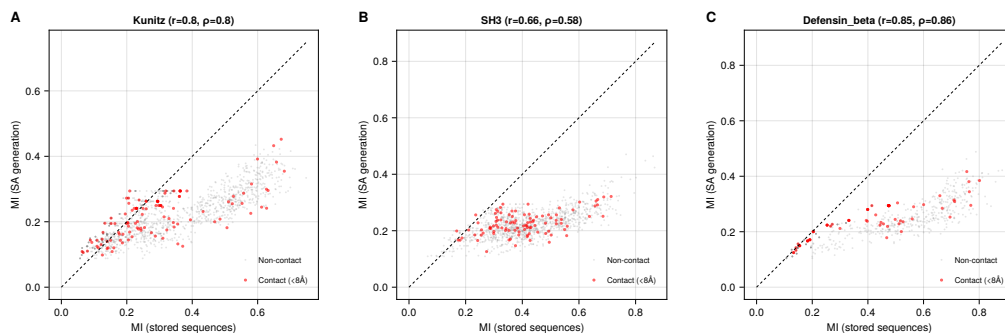


Figure S3: Pairwise mutual information (MI) scatter plots for three representative families. Each point represents one position pair (i, j) ; the x -axis is MI computed from the stored seed alignment and the y -axis is MI from SA-generated sequences. Red points indicate structurally contacting residue pairs ($C\alpha-C\alpha$ distance $< 8 \text{ \AA}$, sequence separation ≥ 5 residues), identified from the family reference PDB structure. The dashed line is the identity. Pearson r and Spearman ρ are shown in each panel title.

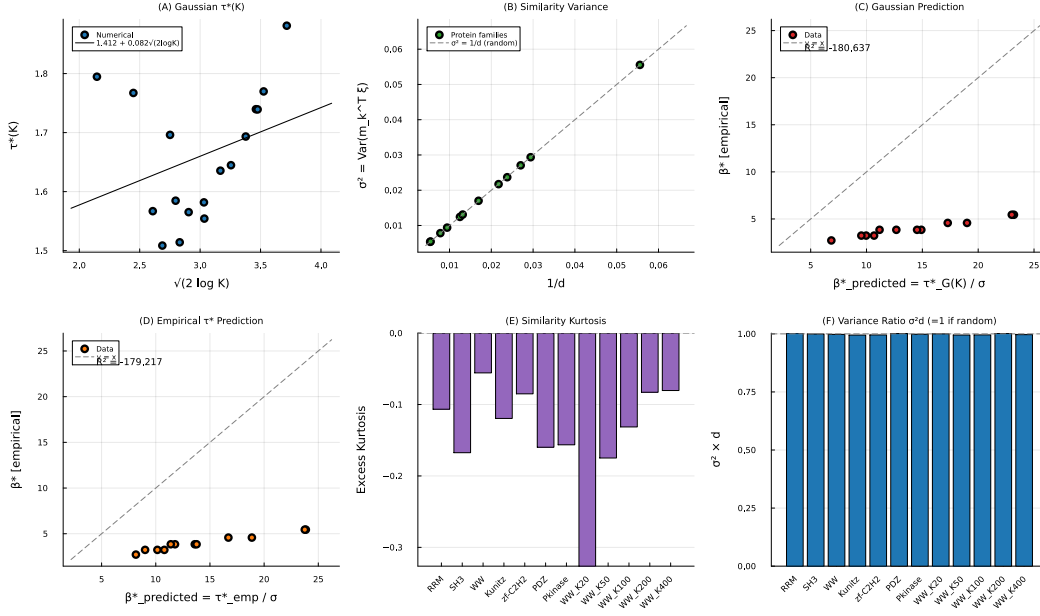


Figure S4: **Analytical investigation of β^* .** (A) Dimensionless softmax inflection $\tau^*(K)$ for Gaussian scores; the dashed line is a linear fit in $\sqrt{2 \log K}$. (B) Similarity variance $\sigma^2 = \text{Var}(\mathbf{m}_k^\top \xi)$ vs. $1/d$ for random probes; all points lie on the identity line, confirming $\sigma^2 = 1/d$. (C) Gaussian prediction $\beta^*_{\text{pred}} = \tau^*_G(K)/\sigma$ vs. empirical β^* ; the prediction has the correct trend ($r = 0.98$) but overshoots by $\sim 4\times$. (D) Same as (C) using the empirical (non-Gaussian) τ^* ; the overshoot persists. (E) Excess kurtosis of the similarity distribution across families. (F) The ratio $\sigma^2 d$ across all datasets, confirming the concentration-of-measure prediction $\sigma^2 d = 1$.

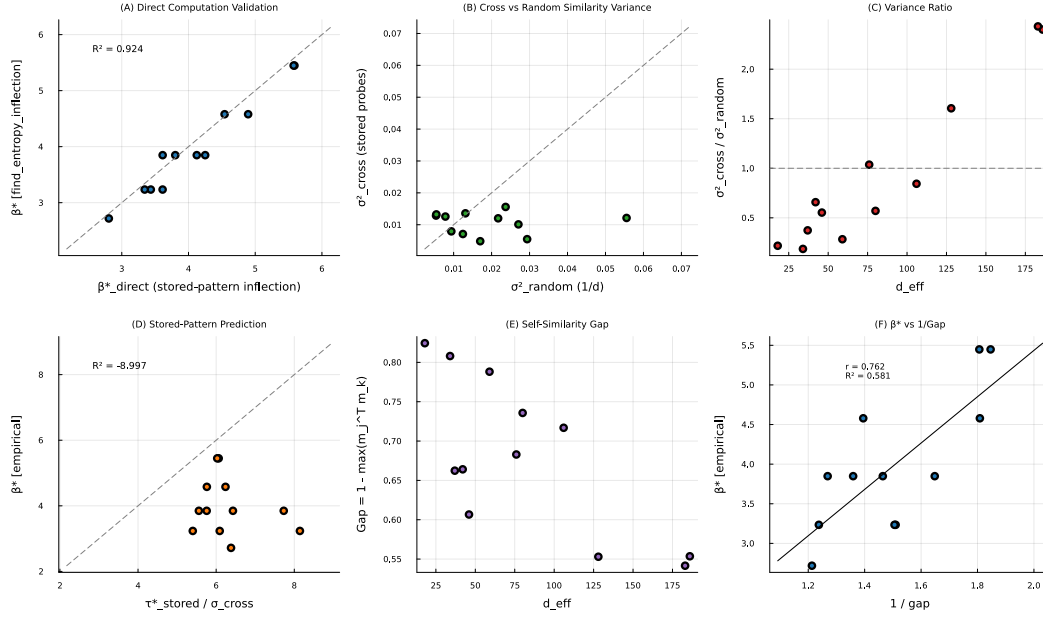


Figure S5: **Stored-pattern probe analysis.** (A) Validation: β^* recomputed from stored-pattern inflection vs. the value from find_entropy_inflection ($R^2 = 0.924$). (B) Cross-similarity variance (stored probes) vs. random-probe variance ($1/d$); stored probes systematically deviate from the identity line. (C) Variance ratio $\sigma^2_{\text{cross}}/\sigma^2_{\text{random}}$ vs. d ; no consistent relationship. (D) The decomposition $\tau^*_{\text{stored}}/\sigma_{\text{cross}}$ fails as a predictor of β^* ($R^2 = -9.0$). (E) Self-similarity gap $\Delta = 1 - \max_{j \neq k} \mathbf{m}_j^T \mathbf{m}_k$ vs. d . (F) β^* vs. $1/\Delta$ ($r = 0.76$); the gap has moderate predictive power but is noisier than $\sqrt{d}$.